notebook

August 28, 2025

1 To do

- ~~Analyze lick data~~
  - ~~RT~~
  - ~~Lick rate~~
  - ~~ILI~~
- History kernels
- Add HMM

2 Analyses of drug data from batch #6

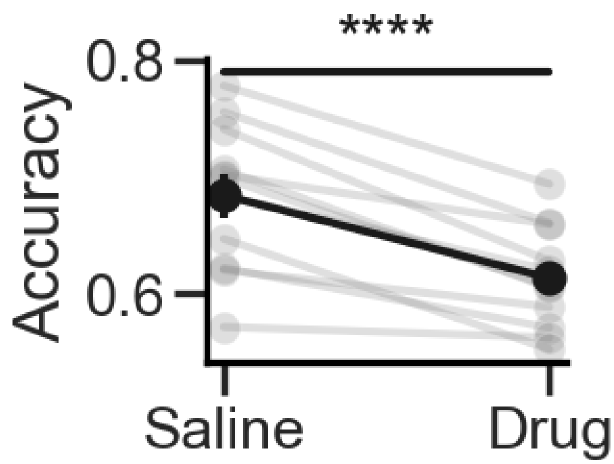
3 Load behavioral data

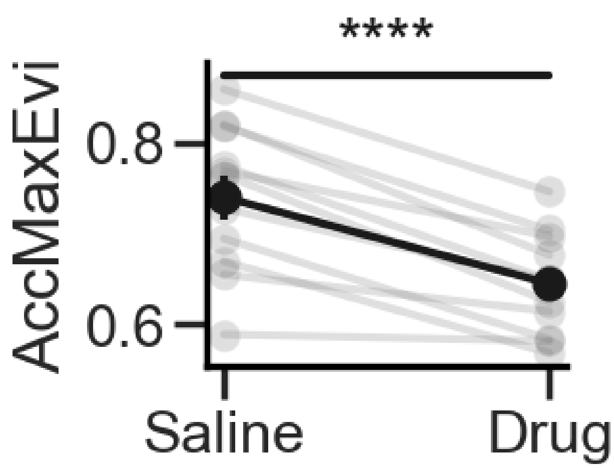
4 Filter data

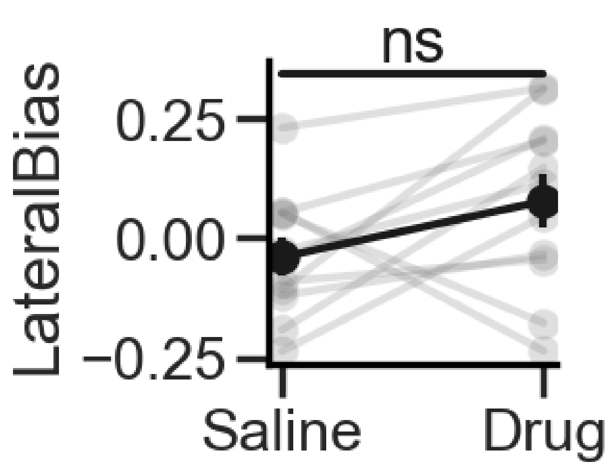
5 Behavioral metrics

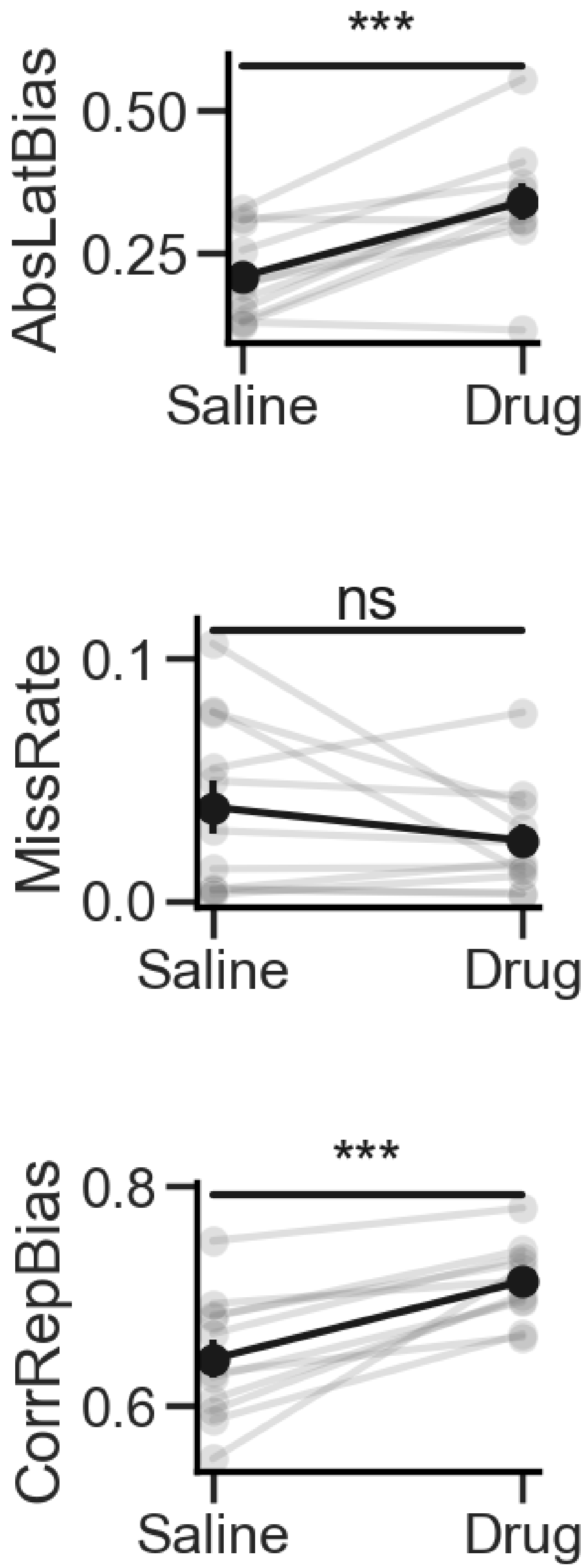
5.1 Standard

```
Accuracy: t-statistic = 7.026, p-value = 0.000
AccMaxEvi: t-statistic = 7.475, p-value = 0.000
LateralBias: t-statistic = -1.768, p-value = 0.107
AbsLatBias: t-statistic = -5.165, p-value = 0.000
MissRate: t-statistic = 1.427, p-value = 0.184
CorrRepBias: t-statistic = -5.479, p-value = 0.000
```


****
Accuracy
0.8
0.6
Saline
Drug


****
AccMaxEvi
0.8
0.6
Saline
Drug


ns
LateralBias
0.25
0.00
−0.25
Saline
Drug

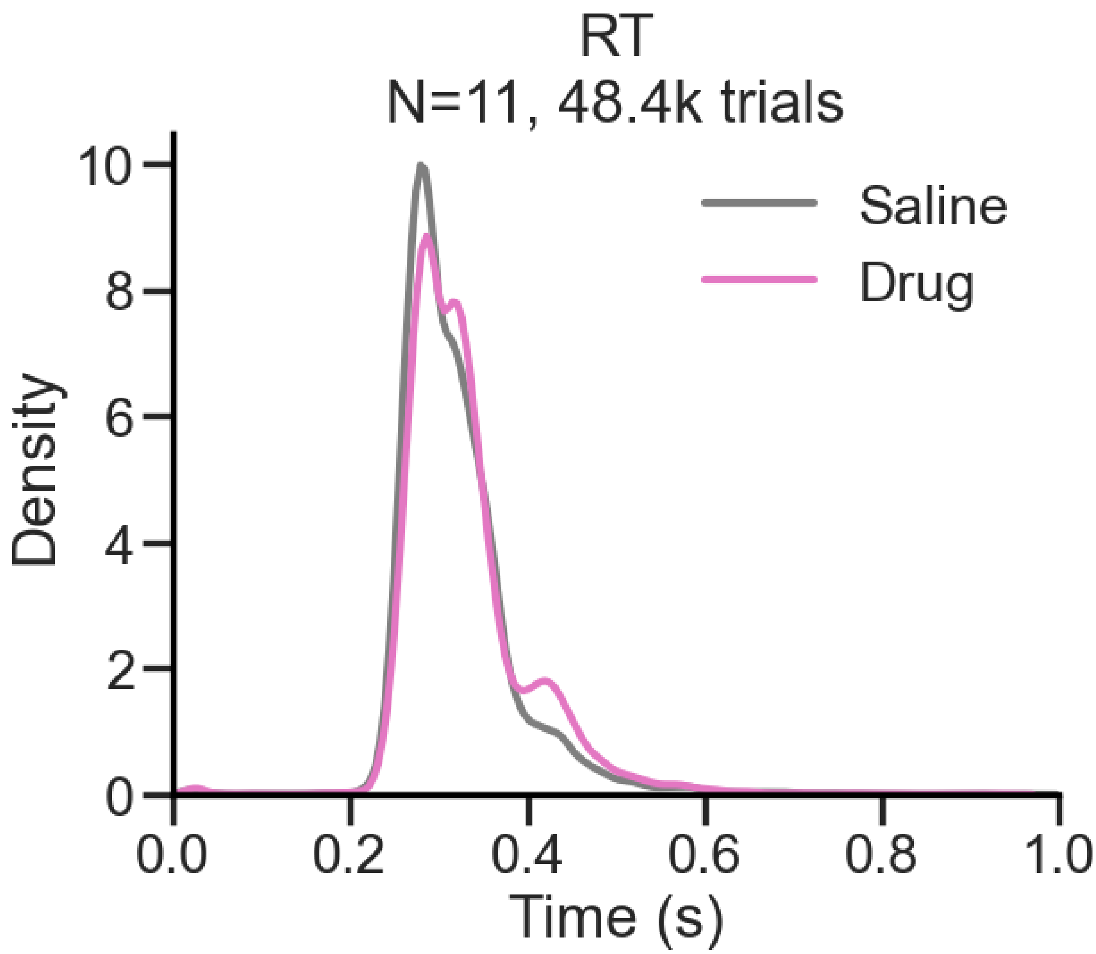

***
AbsLatBias
0.50
0.25
Saline
Drug
ns
MissRate
0.1
0.0
Saline
Drug
***
CorrRepBias
0.8
0.6
Saline
Drug

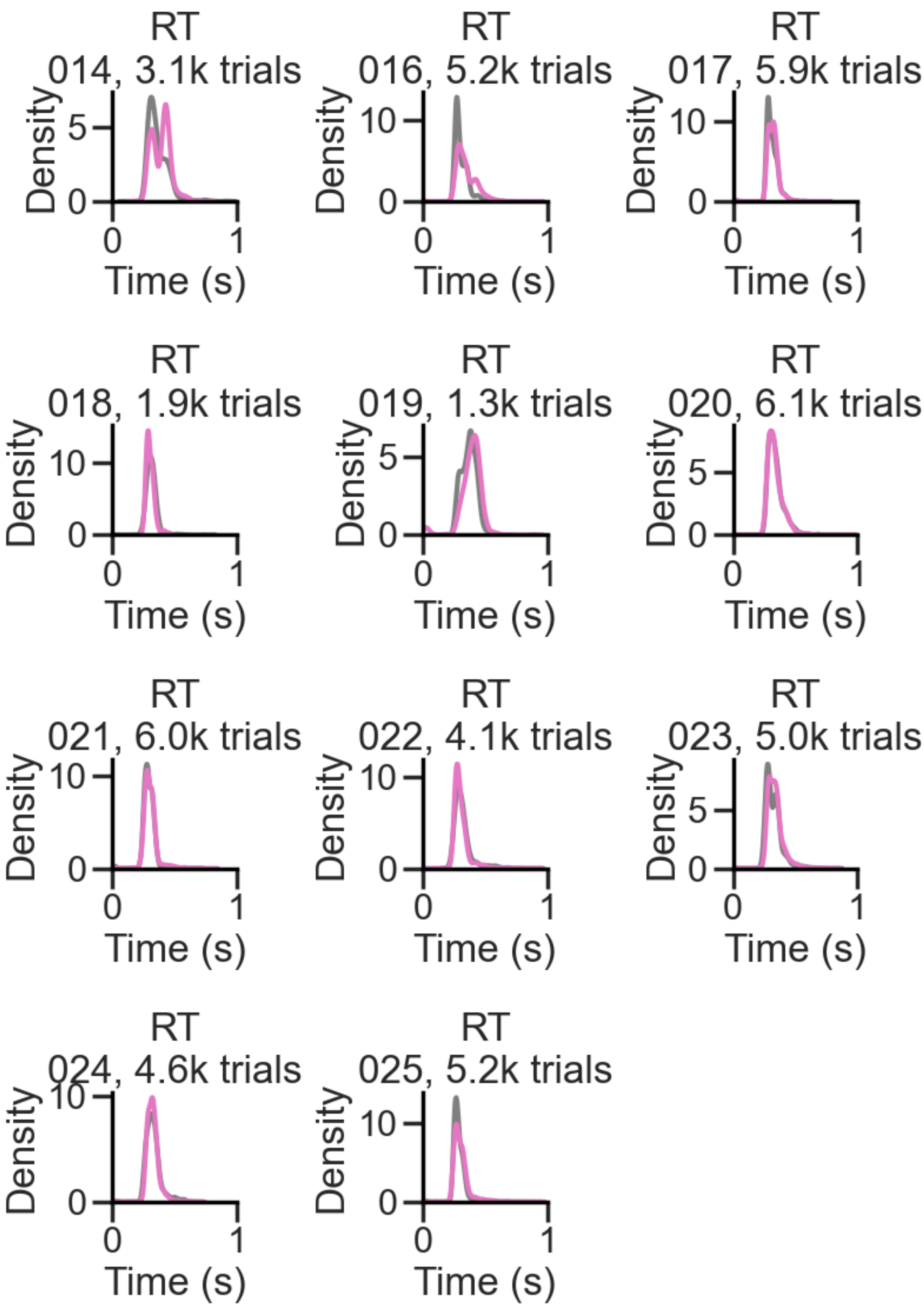
5.2 Licks

5.2.1 Add lick data

```
1.50% of premature lick trials (before response window)
Port1In or Port2In are already lists
['absILD', 'LicksLeft', 'LicksRight', 'nLicksLeft', 'nLicksRight', 'nLicks',
'leftILI', 'rightILI', 'ILI', 'RT', 'RT2', 'SessionHalf']
```

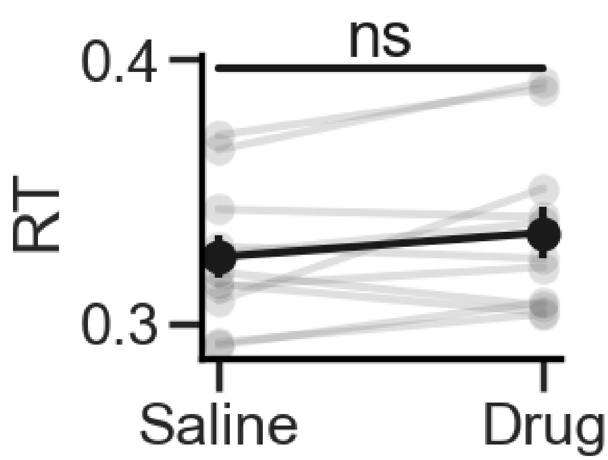
5.2.2 RTs

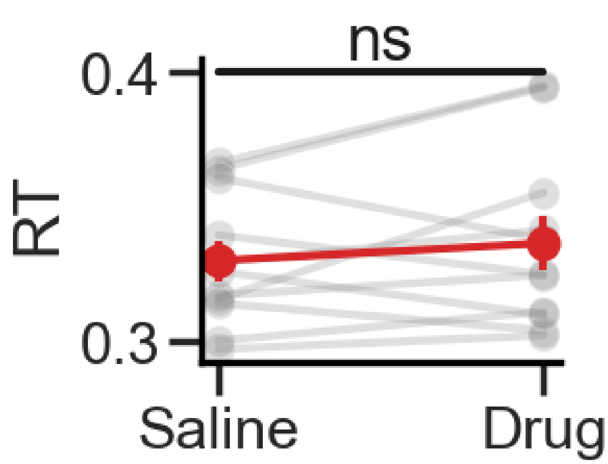


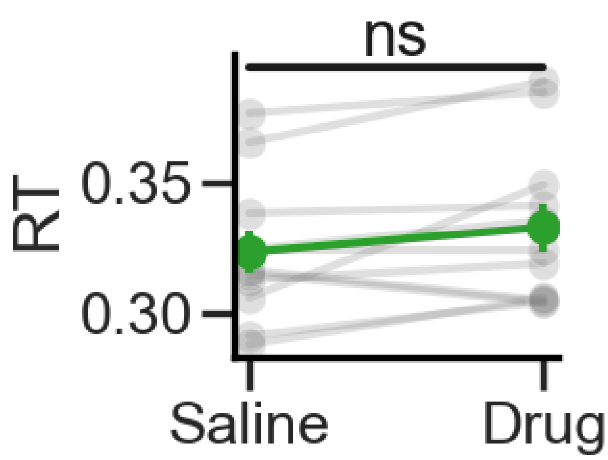



```
RT: t-statistic = -1.848, p-value = 0.094
```

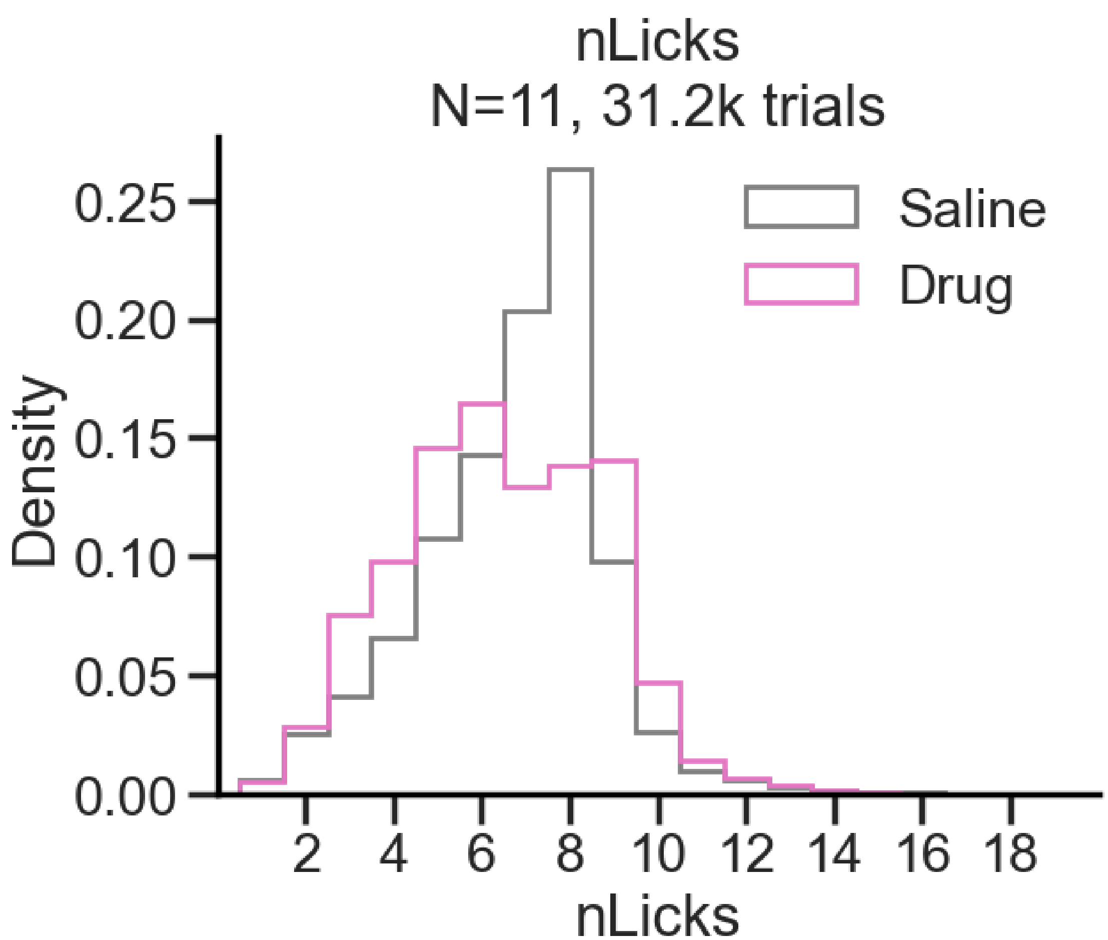
```
RT: t-statistic = -1.027, p-value = 0.328
RT: t-statistic = -2.001, p-value = 0.073
```

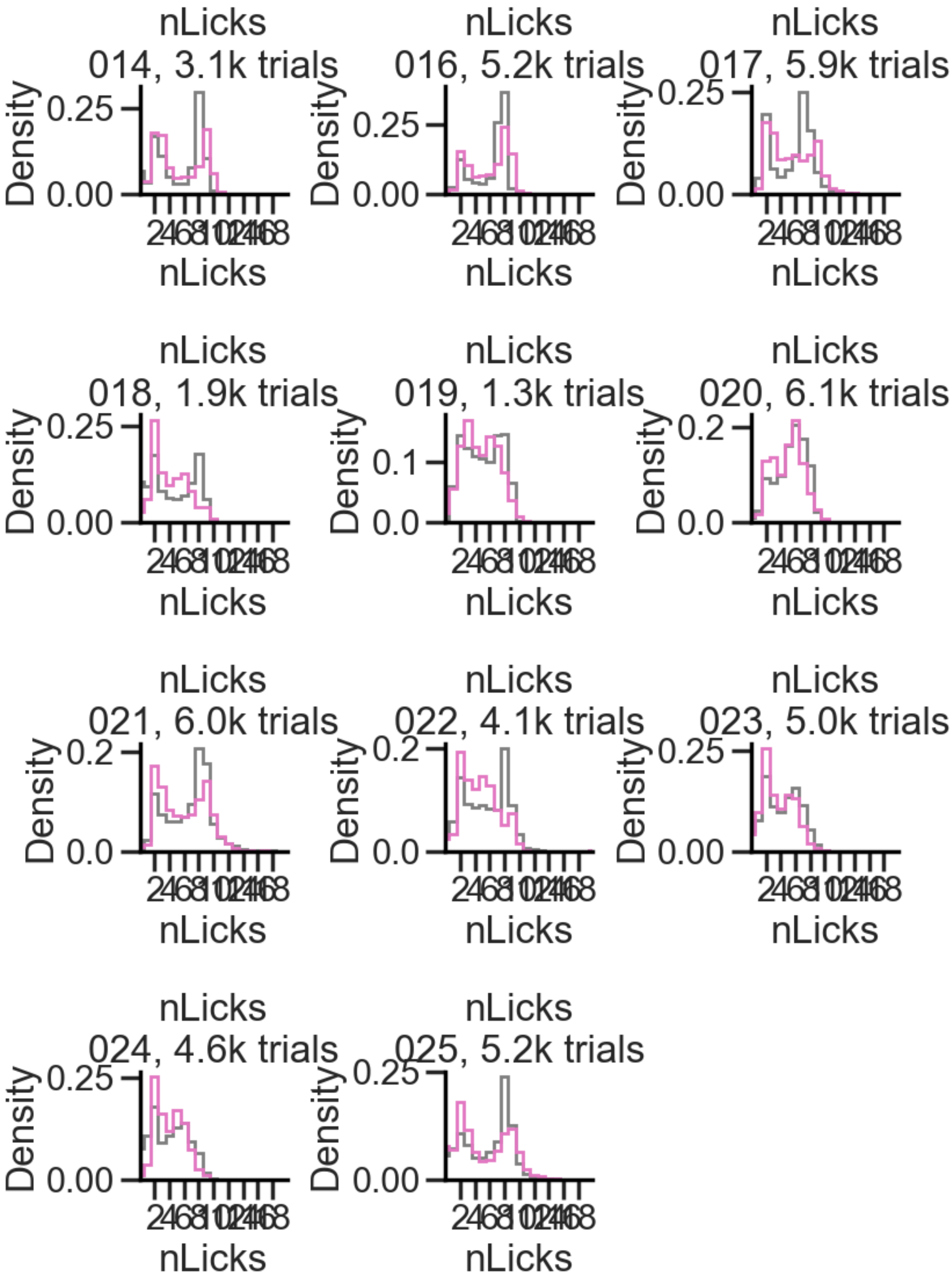




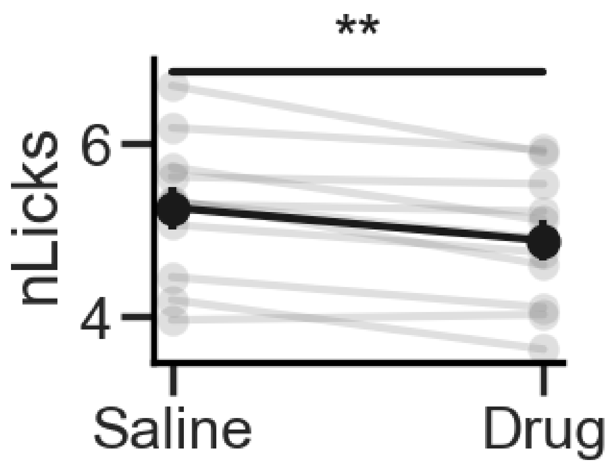


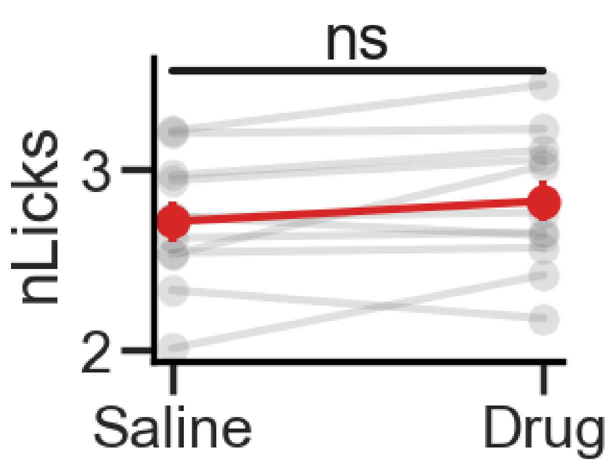
5.2.3 Lick rate

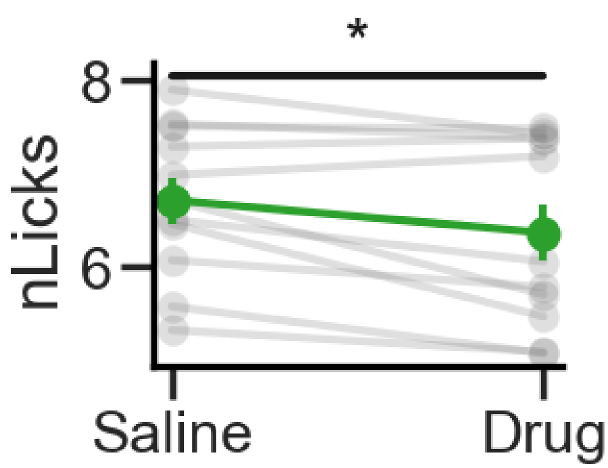




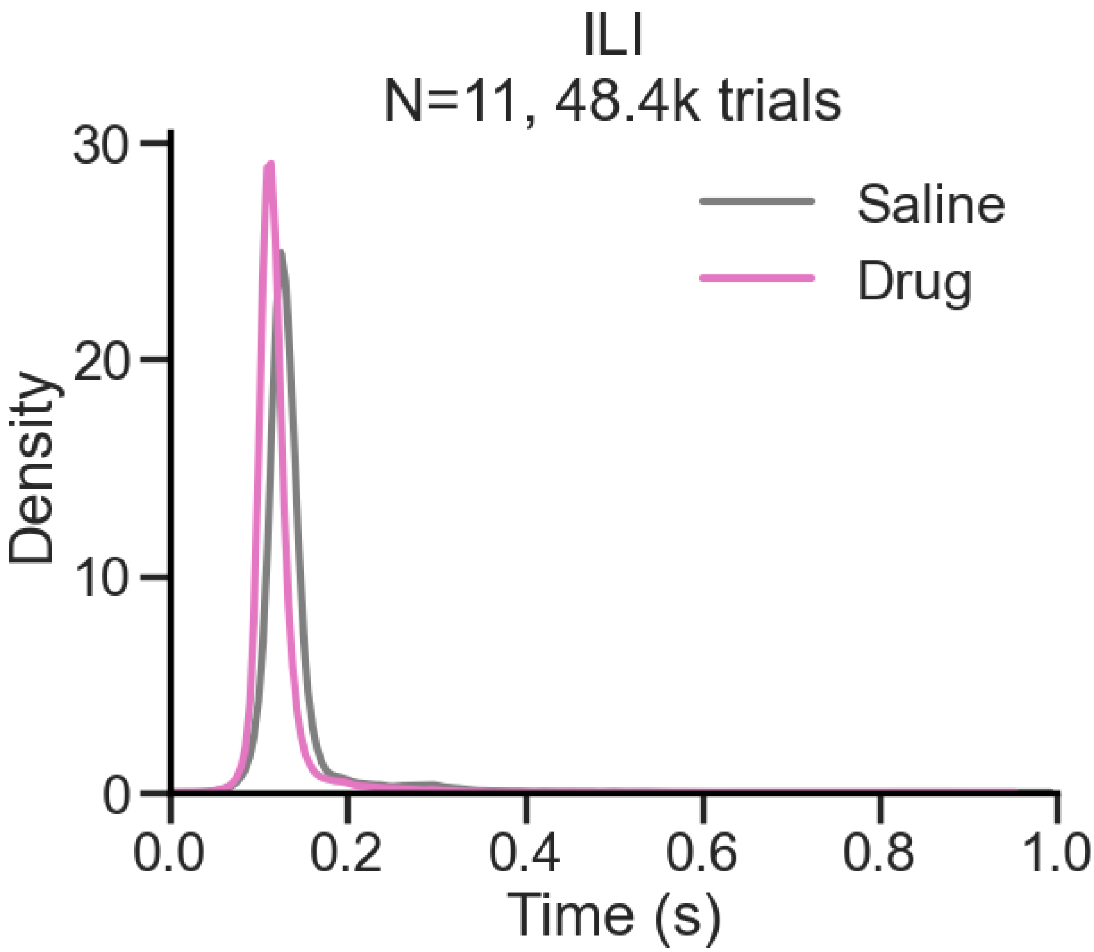

```
nLicks: t-statistic = 4.499, p-value = 0.001
nLicks: t-statistic = -1.904, p-value = 0.086
nLicks: t-statistic = 2.929, p-value = 0.015
```

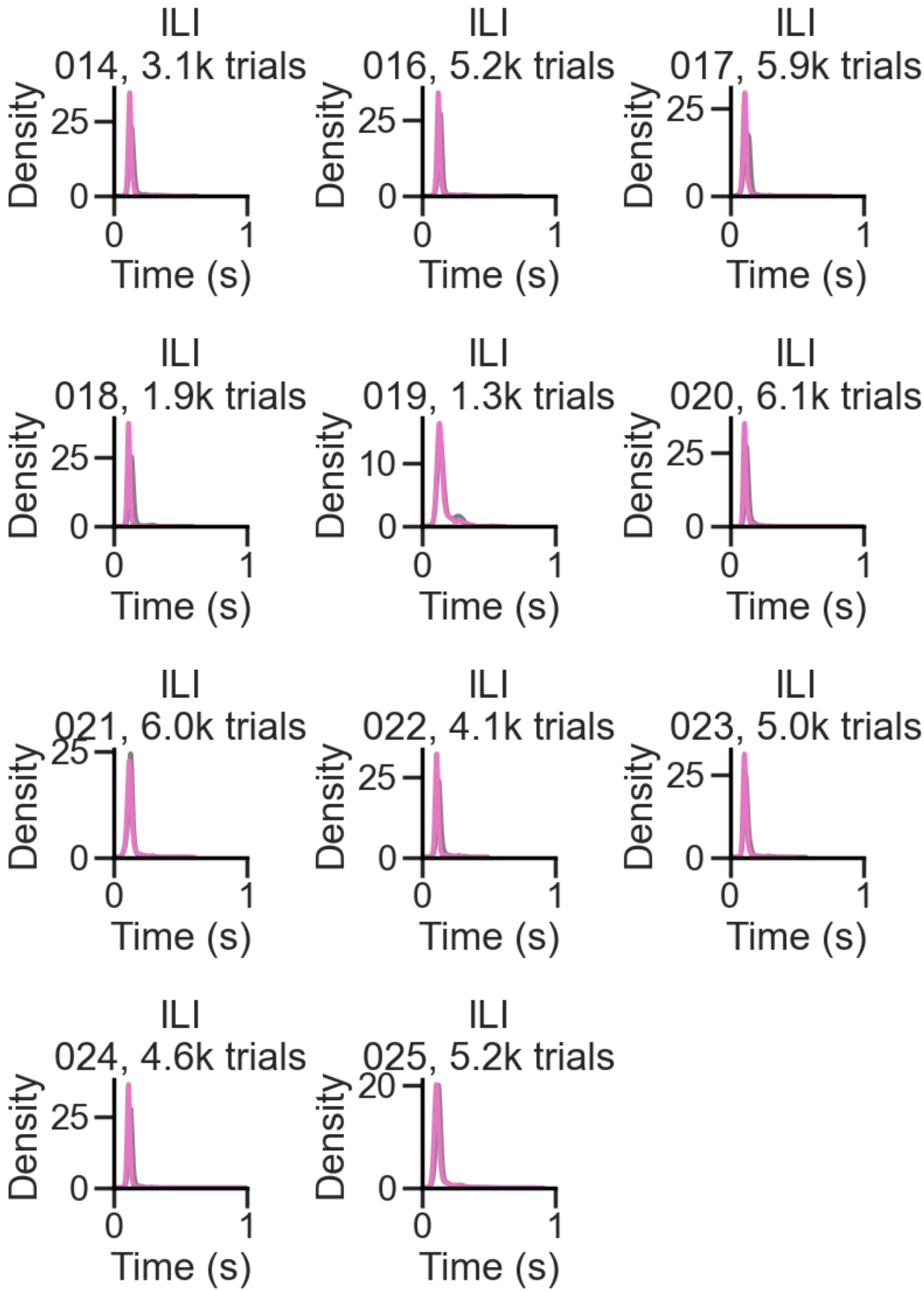

**
nLicks
6
4
Saline
Drug


ns
nLicks
3
2
Saline
Drug


*
nLicks
8
6
Saline
Drug

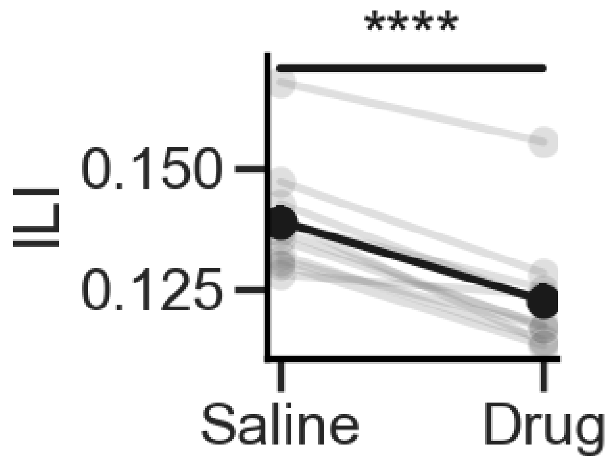
5.2.4 ILI

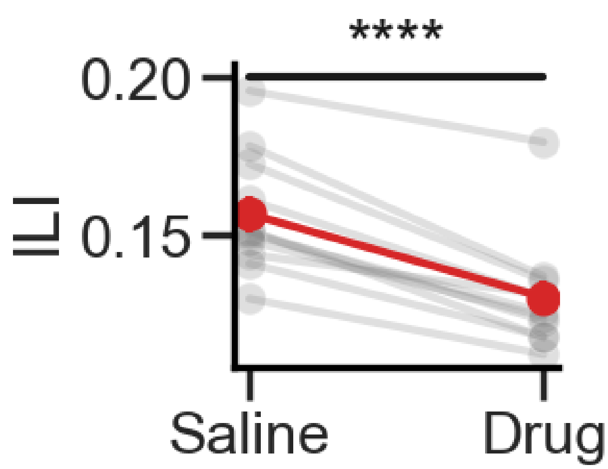

ILI
N=11, 48.4k trials
Saline
Drug
Density
30
20
10
0
0.0
0.2
0.4
0.6
0.8
1.0
Time (s)

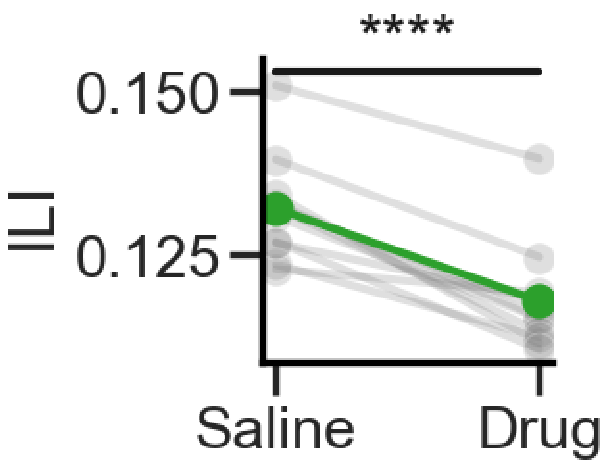



```
ILI: t-statistic = 10.071, p-value = 0.000
```

```
ILI: t-statistic = 9.768, p-value = 0.000
ILI: t-statistic = 8.963, p-value = 0.000
```





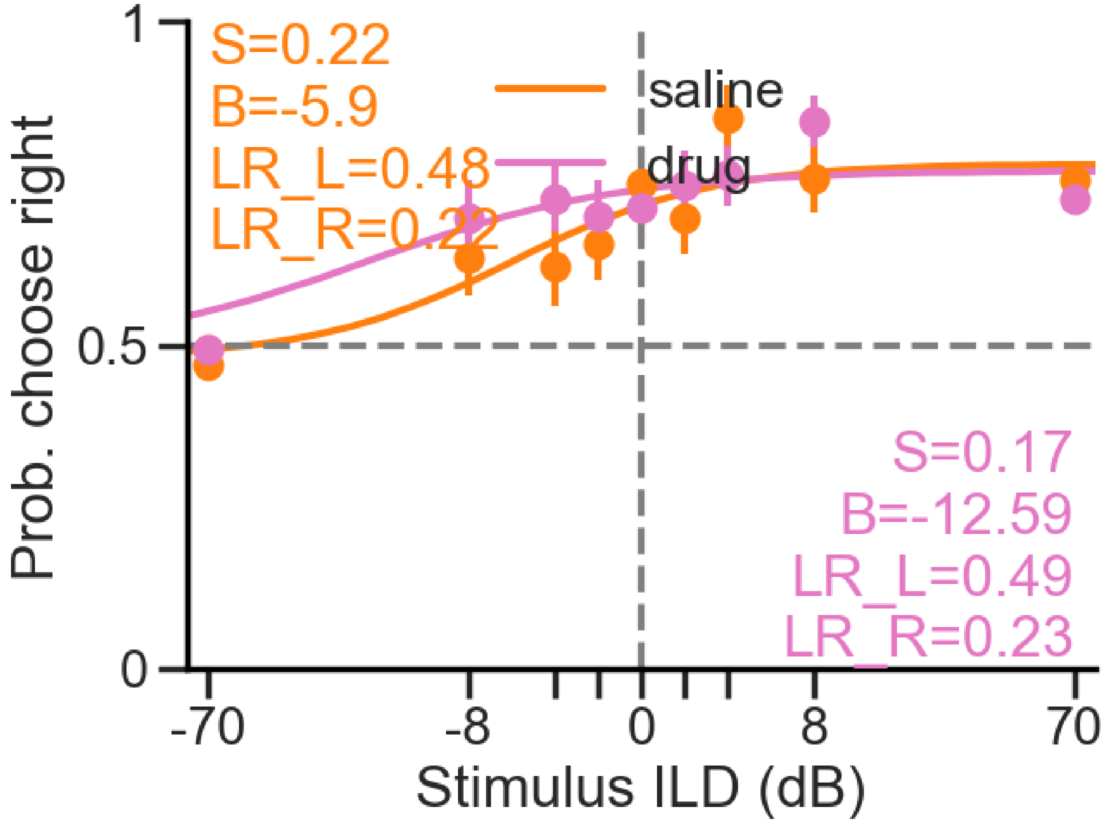


6 Psychometrics

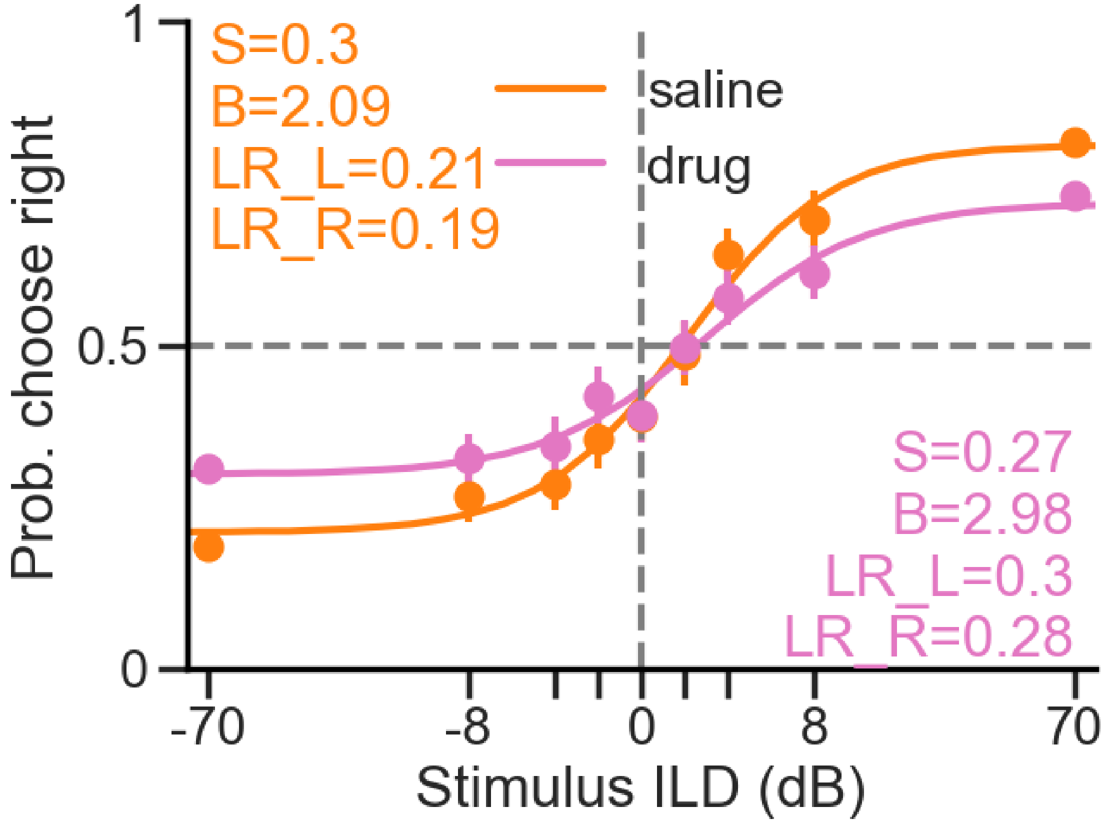
6.1 Probability choose RIGHT

6.1.1 Single animals

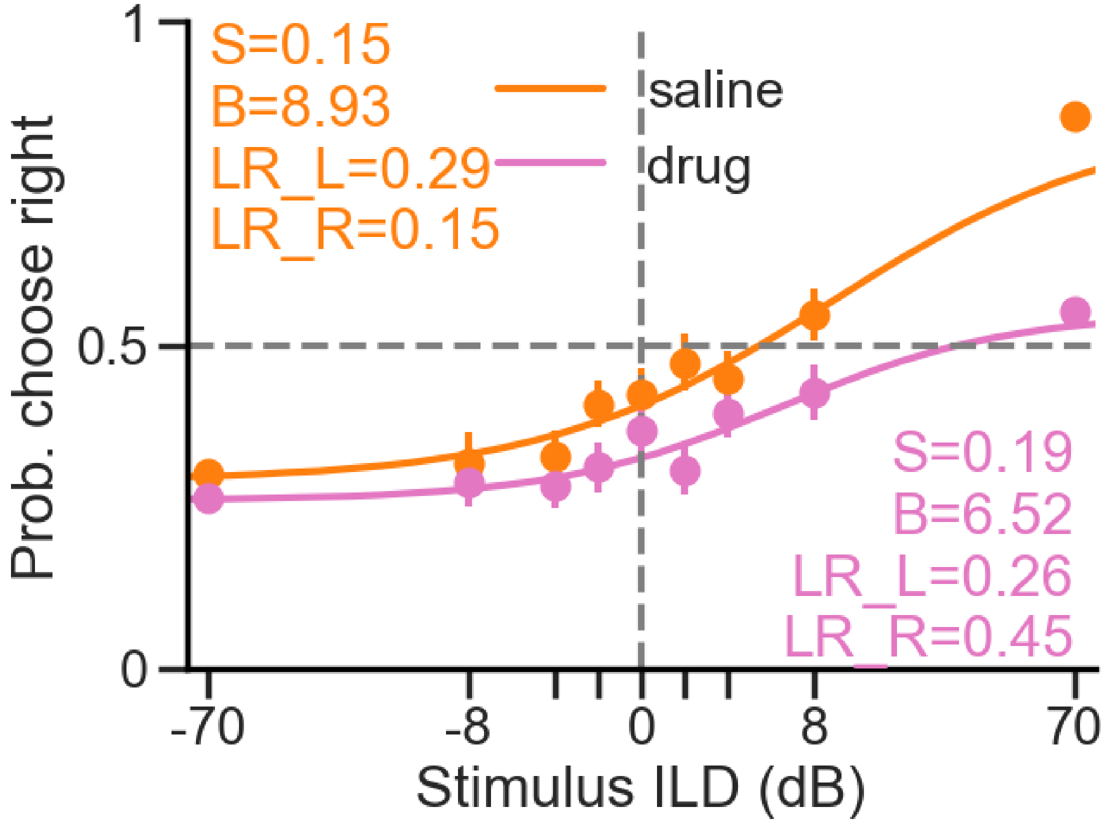
```
Animal 014
Animal 016
Animal 017
Animal 018
Animal 019
Animal 020
Animal 021
Animal 022
Animal 023
Animal 024
Animal 025
```



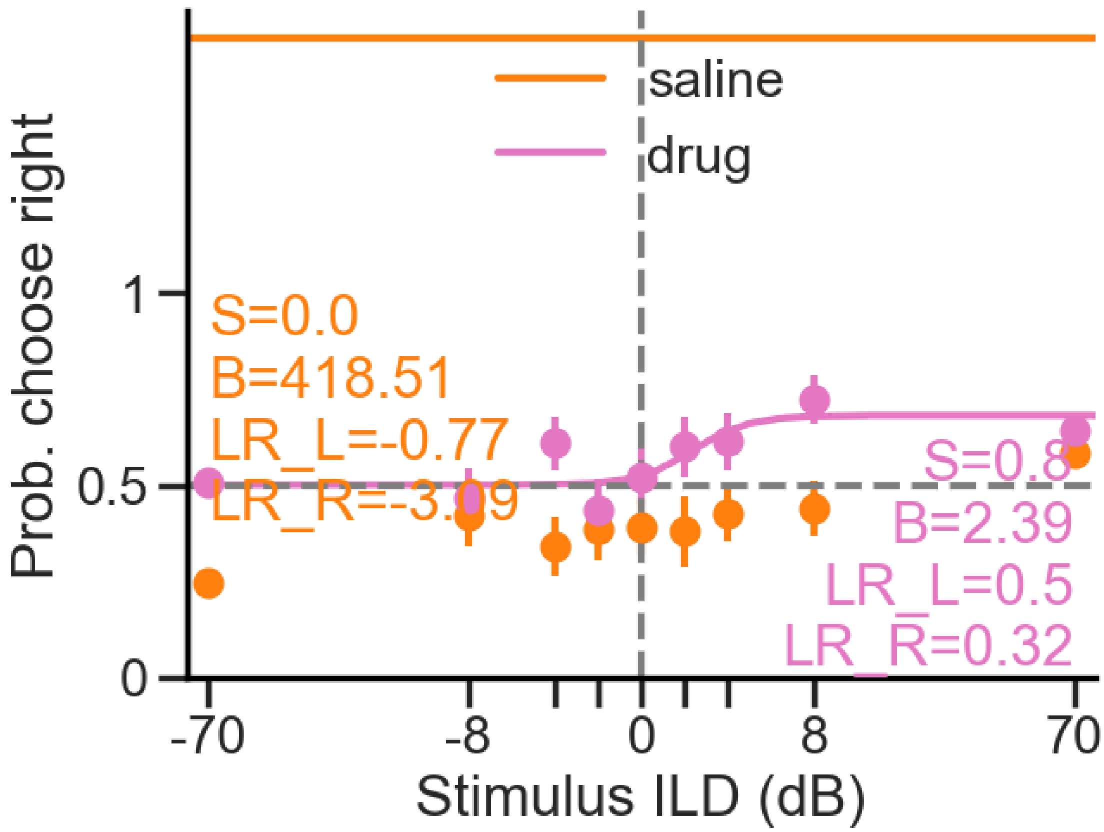
S=0.22
B=-5.9
LR_L=0.48
LR_R=0.22
saline
drug
S=0.17
B=-12.59
LR_L=0.49
LR_R=0.23
Prob. choose right
1
0.5
0
-70
-8
0
8
70
Stimulus ILD (dB)

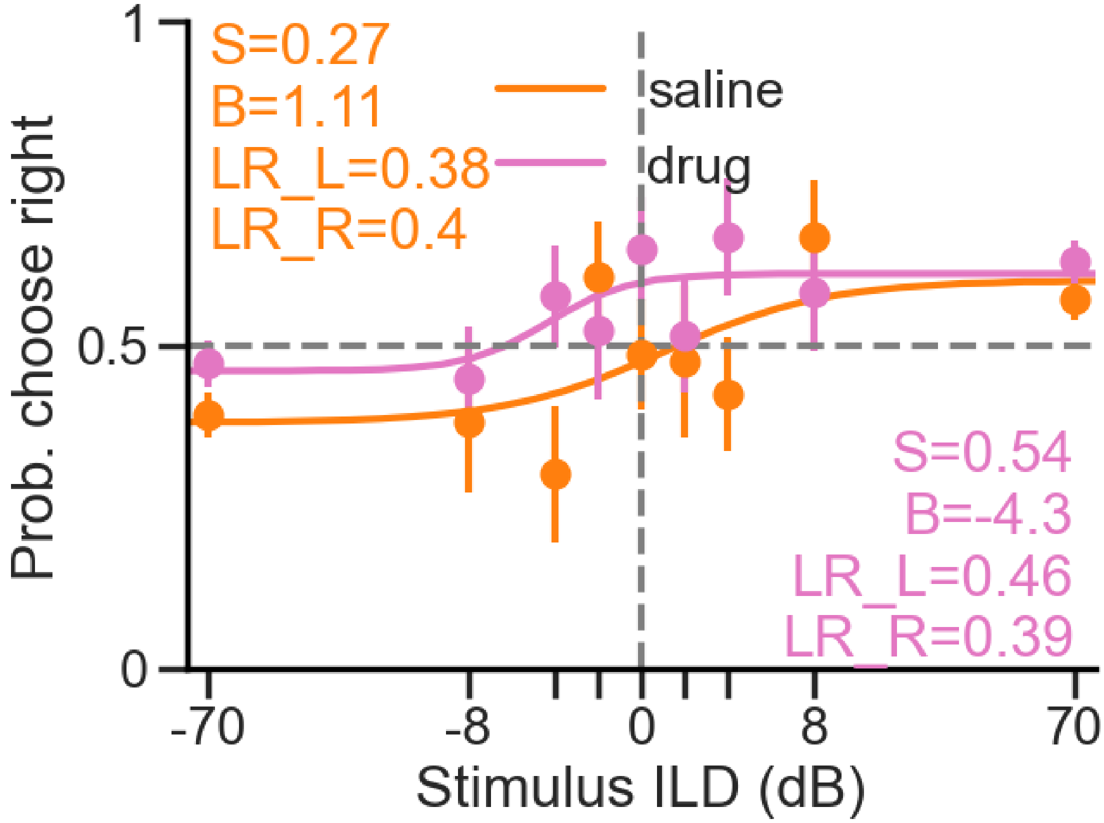


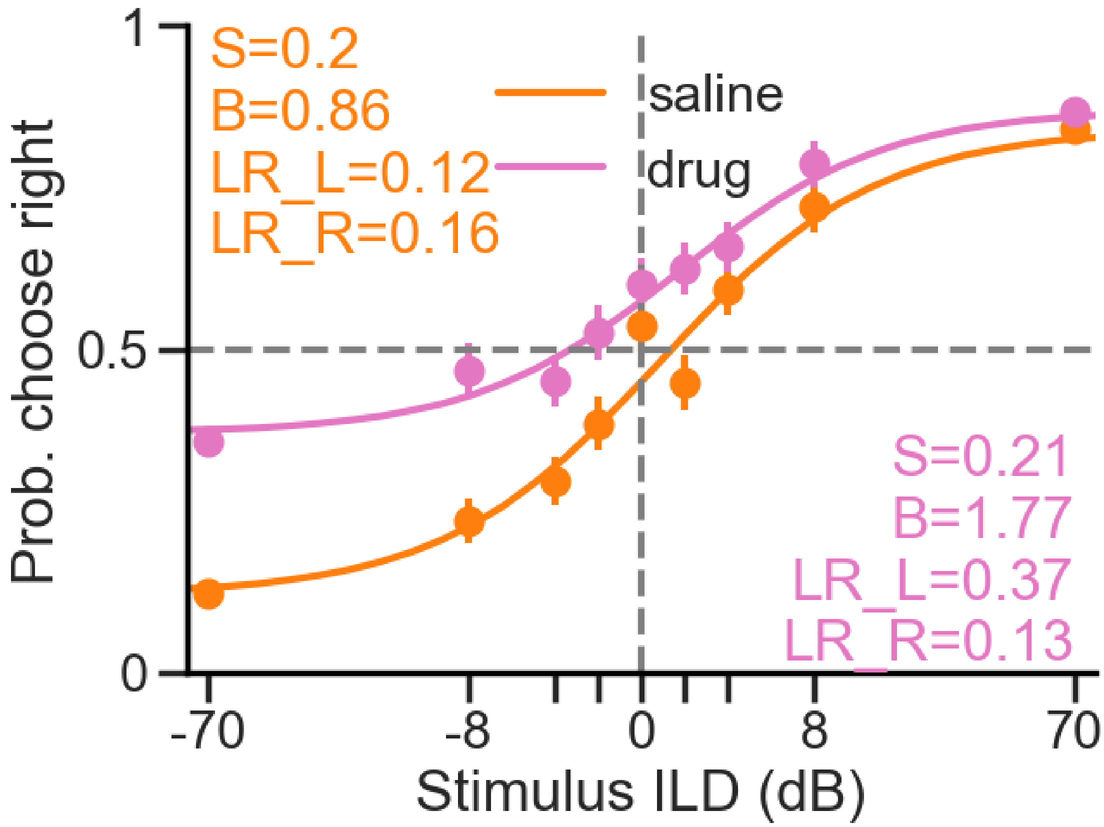
S=0.3
B=2.09
LR_L=0.21
LR_R=0.19
saline
drug
S=0.27
B=2.98
LR_L=0.3
LR_R=0.28
1
0.5
0
Prob. choose right
-70
-8
0
8
70
Stimulus ILD (dB)

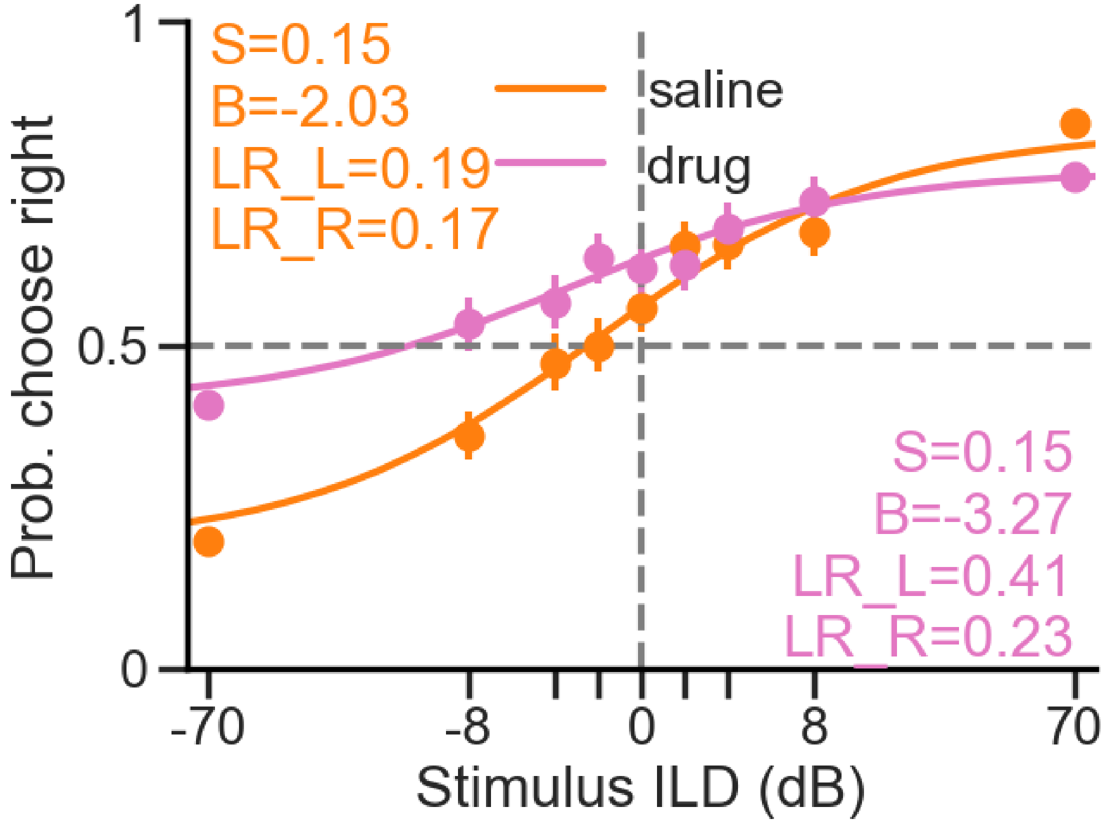


1
0.5
0
S=0.15
B=8.93
LR_L=0.29
LR_R=0.15
saline
drug
S=0.19
B=6.52
LR_L=0.26
LR_R=0.45
Prob. choose right
-70
-8
0
8
70
Stimulus ILD (dB)

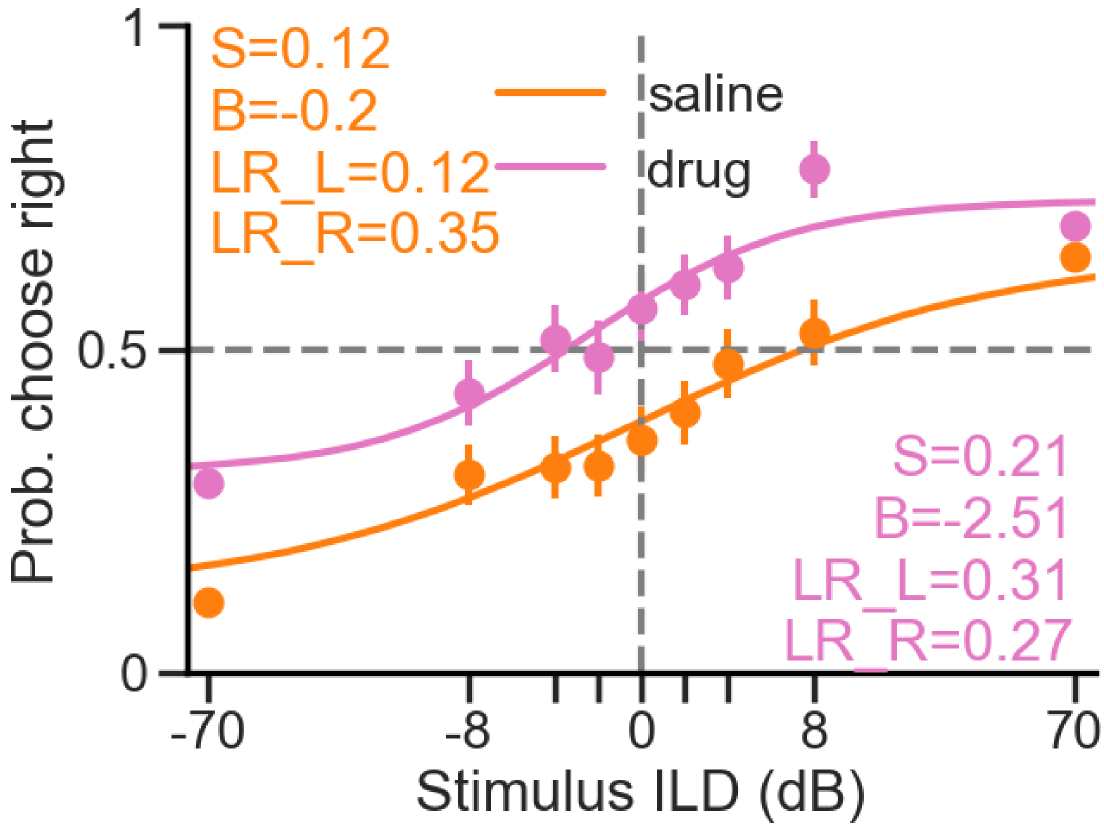

saline
drug
Prob. choose right
1
0.5
0
S=0.0
B=418.51
LR_L=-0.77
LR_R=-3.09
S=0.8
B=2.39
LR_L=0.5
LR_R=0.32
-70
-8
0
8
70
Stimulus ILD (dB)


S=0.27
B=1.11
LR_L=0.38
LR_R=0.4
saline
drug
S=0.54
B=-4.3
LR_L=0.46
LR_R=0.39
Prob. choose right
1
0.5
0
-70
-8
0
8
70
Stimulus ILD (dB)

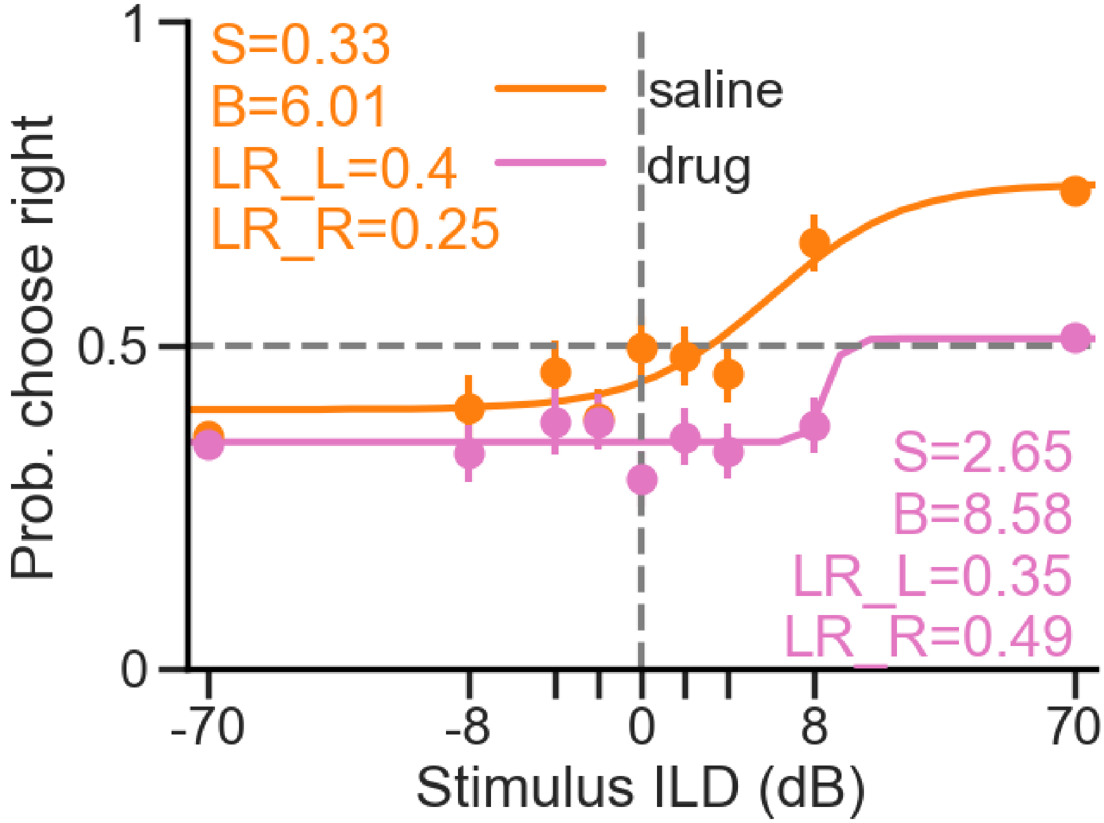

S=0.2
B=0.86
LR_L=0.12
LR_R=0.16
saline
drug
S=0.21
B=1.77
LR_L=0.37
LR_R=0.13
Prob. choose right
1
0.5
0
-70
-8
0
8
70
Stimulus ILD (dB)

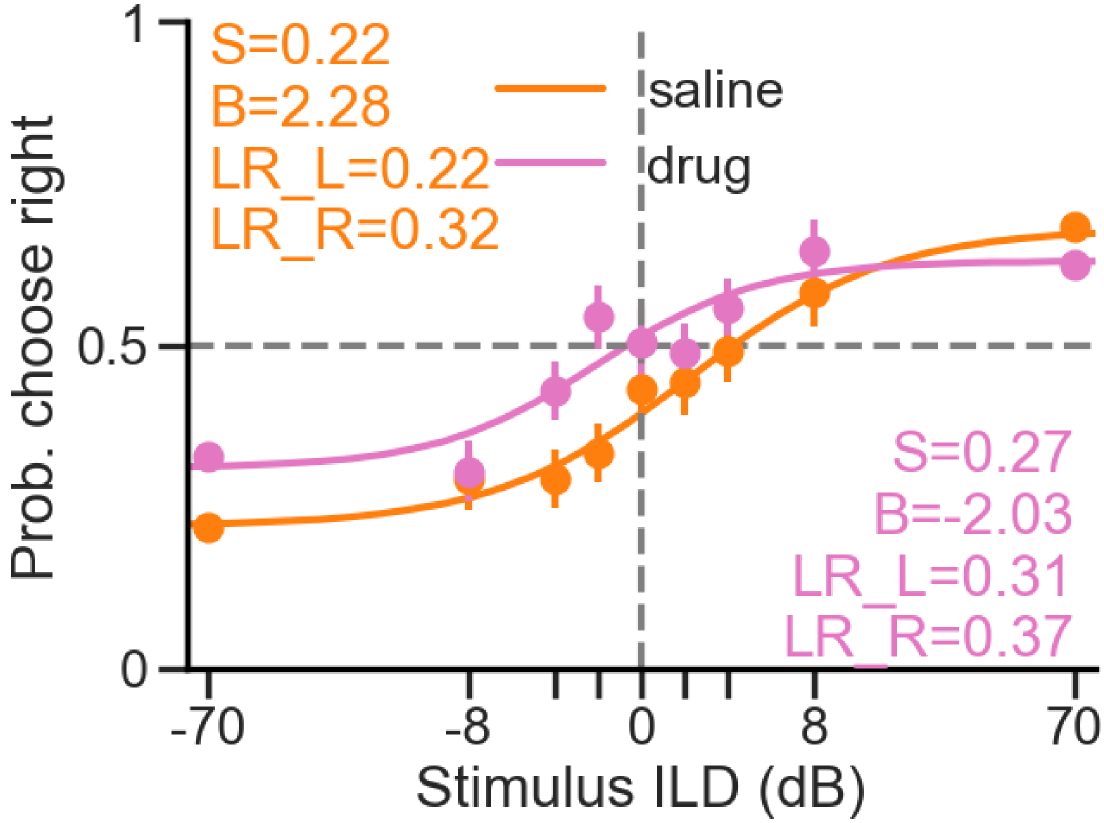


S=0.15
B=-2.03
LR_L=0.19
LR_R=0.17
saline
drug
S=0.15
B=-3.27
LR_L=0.41
LR_R=0.23
Prob. choose right
1
0.5
0
-70
-8
0
8
70
Stimulus ILD (dB)

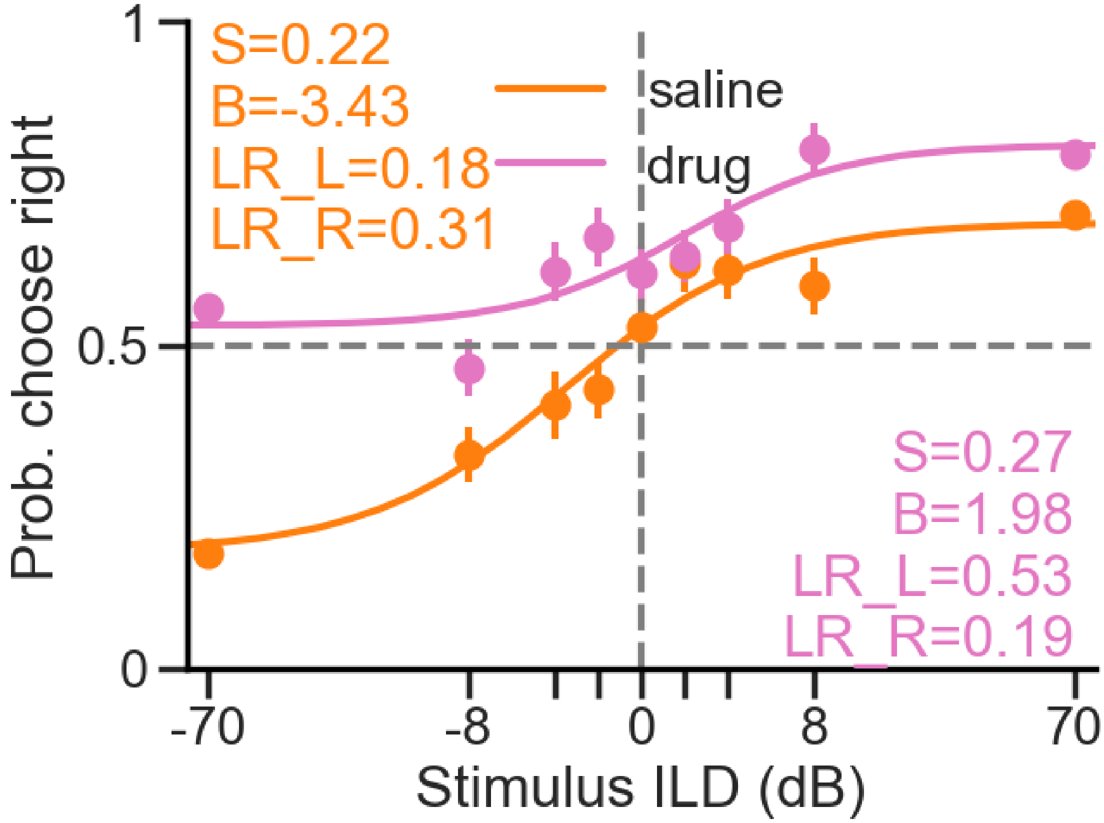


S=0.12
B=-0.2
LR_L=0.12
LR_R=0.35
saline
drug
S=0.21
B=-2.51
LR_L=0.31
LR_R=0.27
Prob. choose right
1
0.5
0
-70
-8
0
8
70
Stimulus ILD (dB)


S=0.33
B=6.01
LR_L=0.4
LR_R=0.25
saline
drug
S=2.65
B=8.58
LR_L=0.35
LR_R=0.49
Prob. choose right
1
0.5
0
-70
-8
0
8
70
Stimulus ILD (dB)

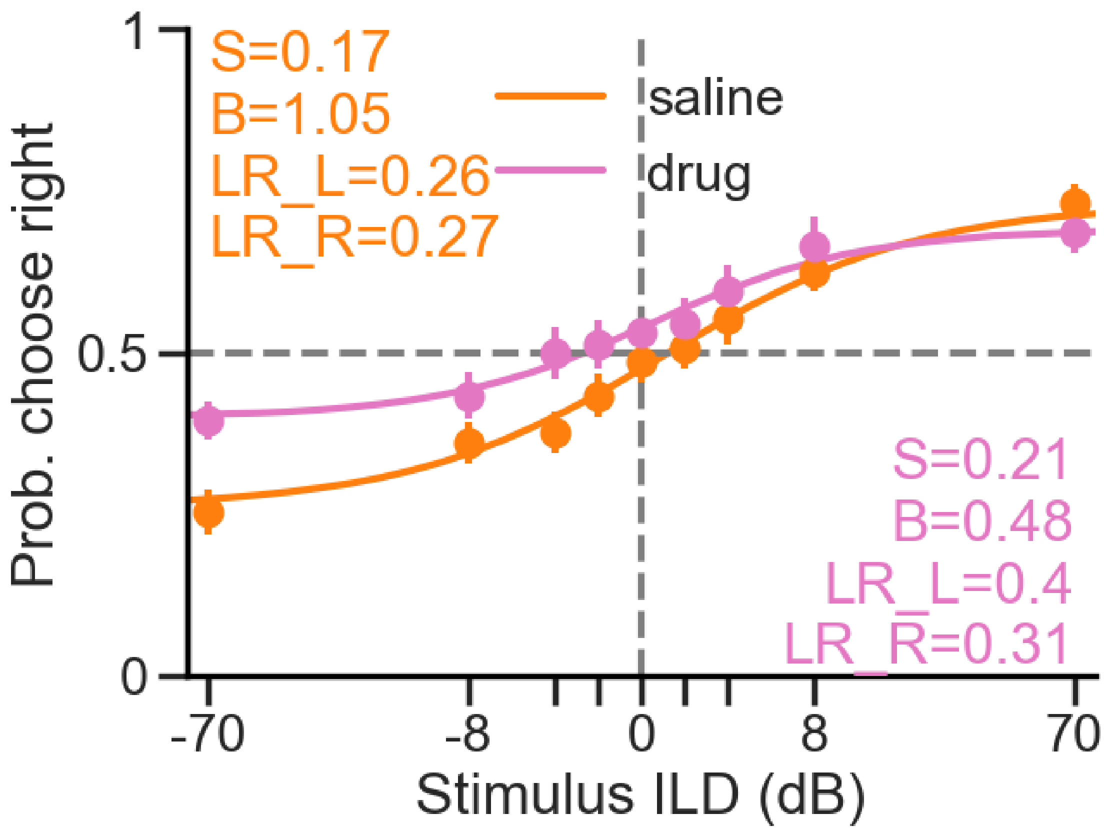


S=0.22
B=2.28
LR_L=0.22
LR_R=0.32
saline
drug
S=0.27
B=-2.03
LR_L=0.31
LR_R=0.37
1
0.5
0
Prob. choose right
-70
-8
0
8
70
Stimulus ILD (dB)



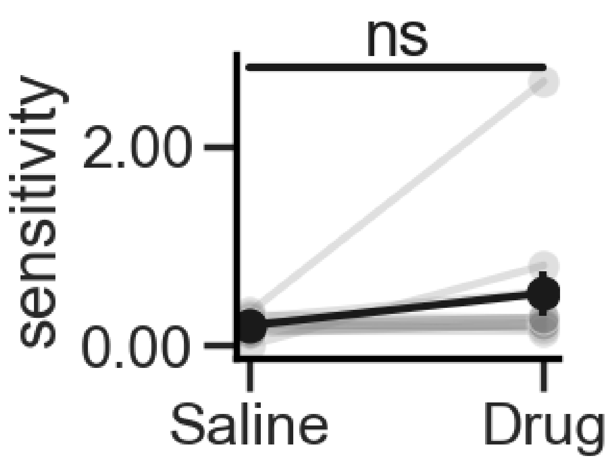
S=0.22
B=-3.43
LR_L=0.18
LR_R=0.31
saline
drug
S=0.27
B=1.98
LR_L=0.53
LR_R=0.19
Prob. choose right
1
0.5
0
-70
-8
0
8
70
Stimulus ILD (dB)

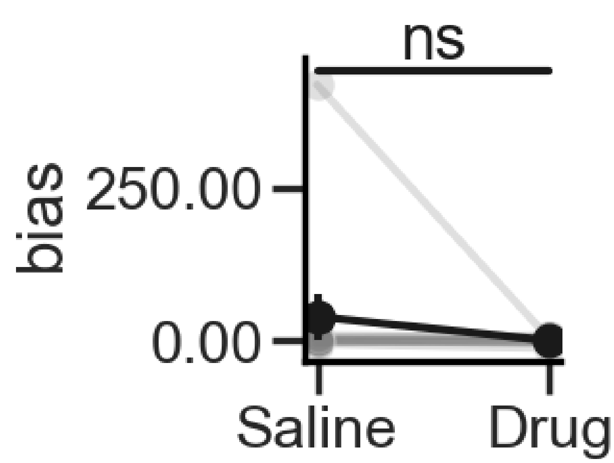
6.1.2 Mean across animals

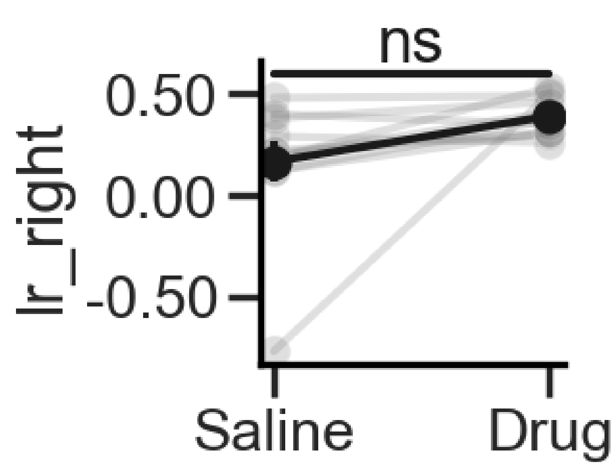


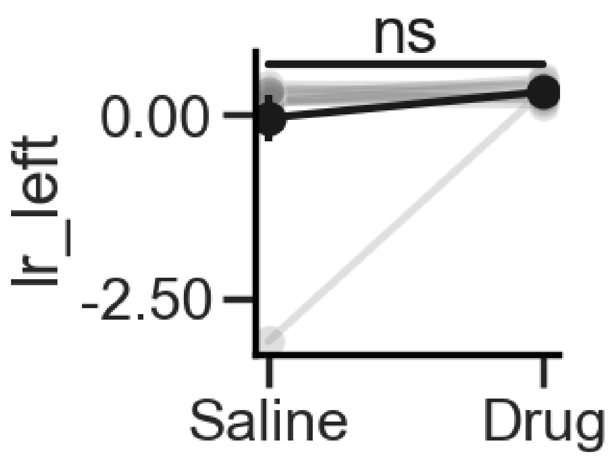
Stats

```
sensitivity: t-statistic = -1.519, p-value = 0.160
bias: t-statistic = 1.033, p-value = 0.326
lr_right: t-statistic = -2.024, p-value = 0.071
lr_left: t-statistic = -1.158, p-value = 0.274
```


ns
sensitivity
2.00
0.00
Saline
Drug


ns
bias
250.00
0.00
Saline
Drug

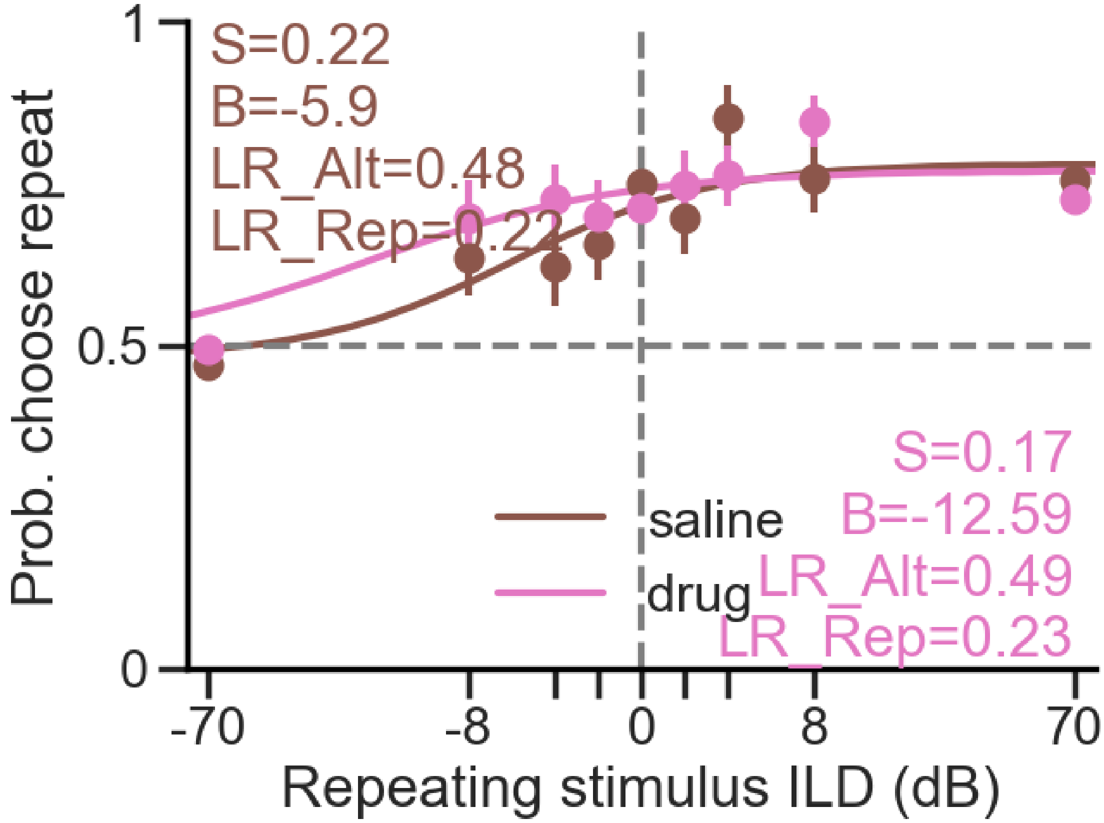

ns
lr_right
0.50
0.00
-0.50
Saline
Drug

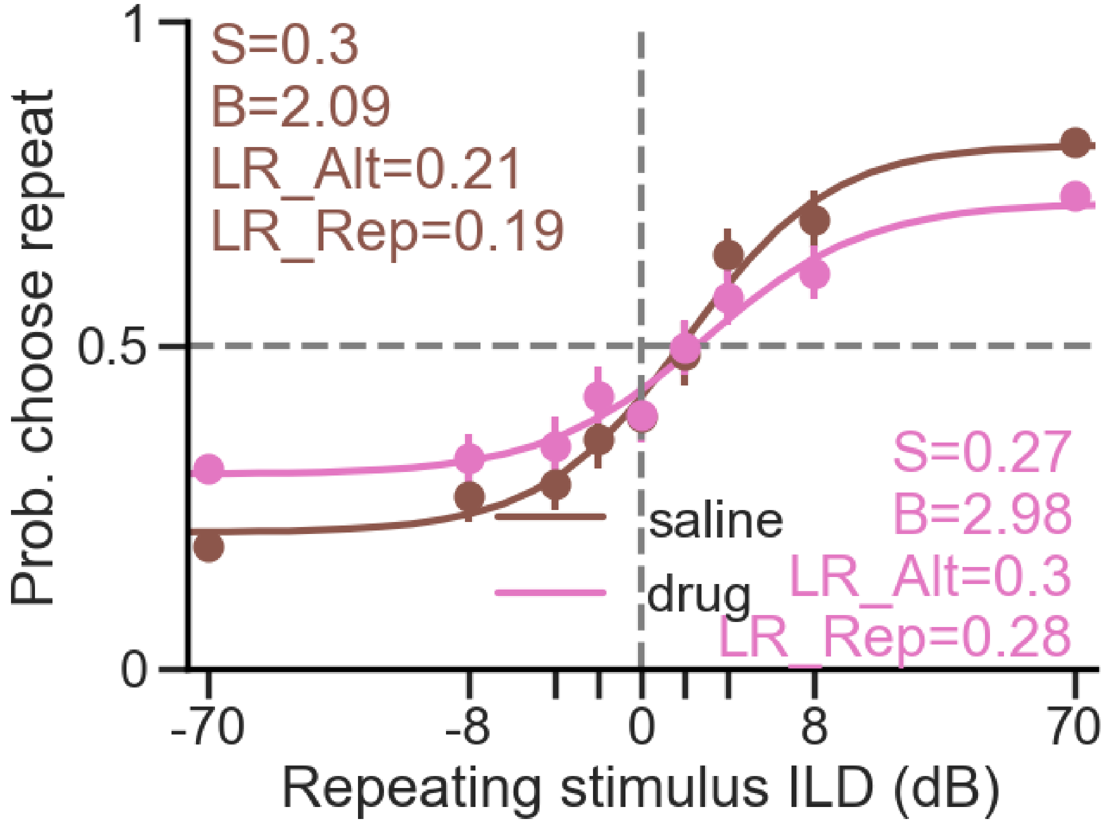


6.2 Probability choose REPEAT

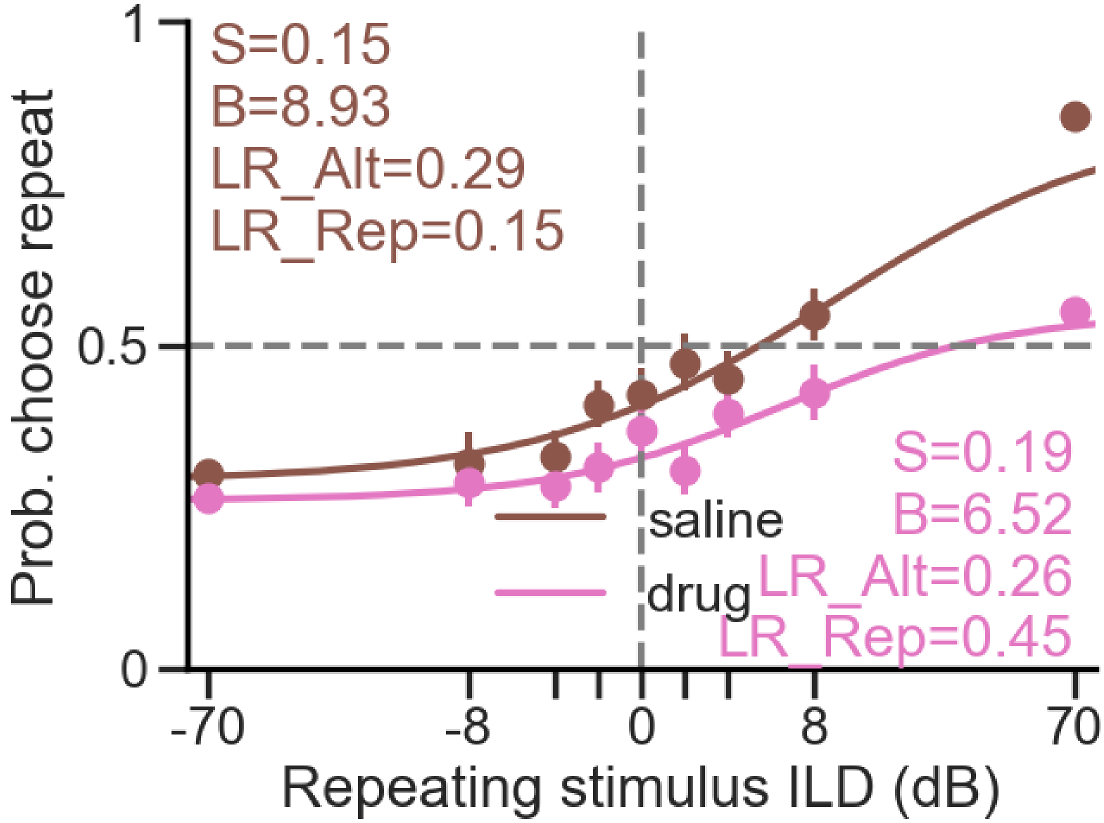
6.2.1 Single animals

```
Animal 014
Animal 016
Animal 017
Animal 018
Animal 019
Animal 020
Animal 021
Animal 022
Animal 023
Animal 024
Animal 025
```

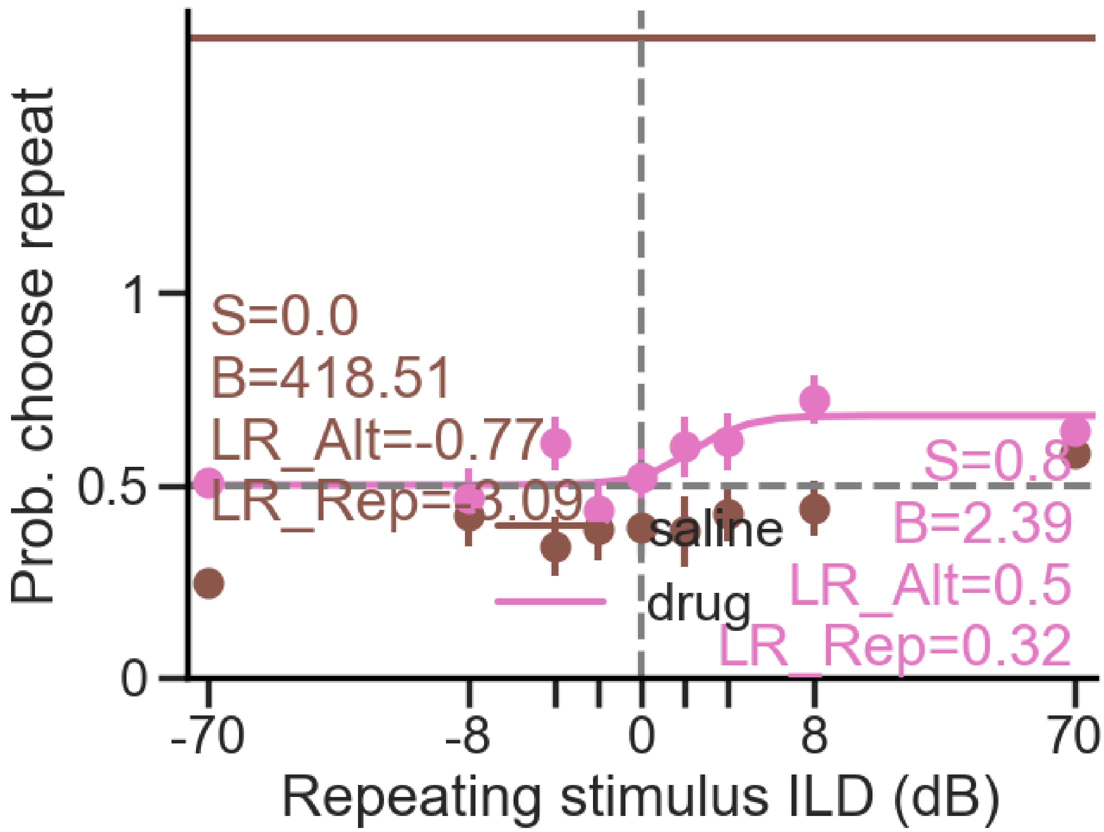

S=0.22
B=-5.9
LR_Alt=0.48
LR_Rep=0.22
S=0.17
B=-12.59
LR_Alt=0.49
LR_Rep=0.23
saline
drug
Prob. choose repeat
1
0.5
0
-70
-8
0
8
70
Repeating stimulus ILD (dB)



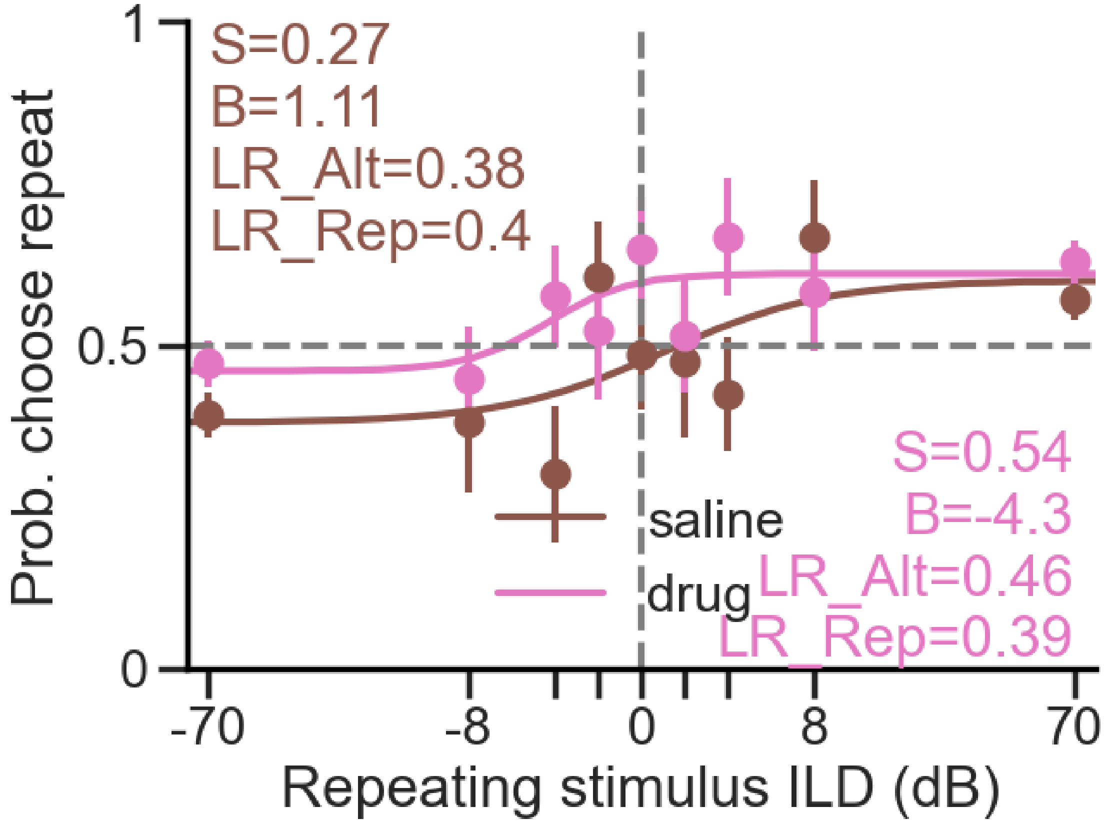
S=0.3
B=2.09
LR_Alt=0.21
LR_Rep=0.19
S=0.27
B=2.98
LR_Alt=0.3
LR_Rep=0.28
saline
drug
Prob. choose repeat
1
0.5
0
-70
-8
0
8
70
Repeating stimulus ILD (dB)



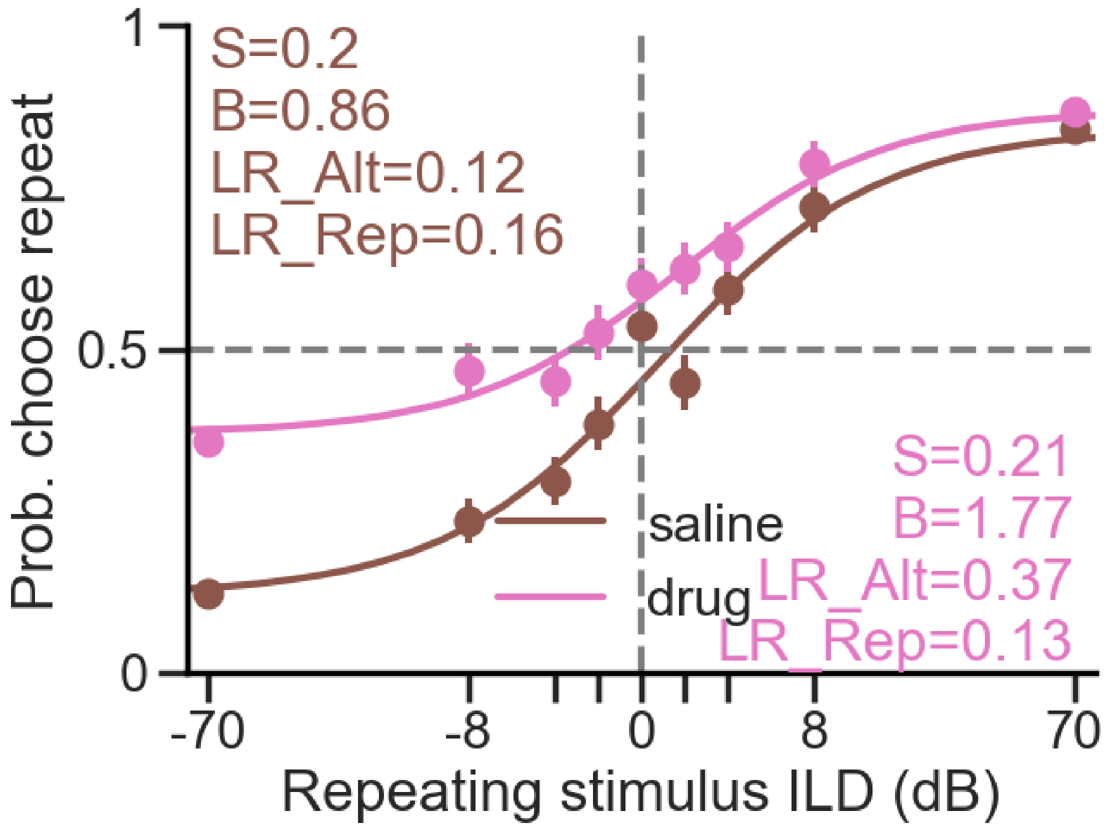
S=0.15
B=8.93
LR_Alt=0.29
LR_Rep=0.15
S=0.19
B=6.52
LR_Alt=0.26
LR_Rep=0.45
saline
drug
Prob. choose repeat
1
0.5
0
-70
-8
0
8
70
Repeating stimulus ILD (dB)

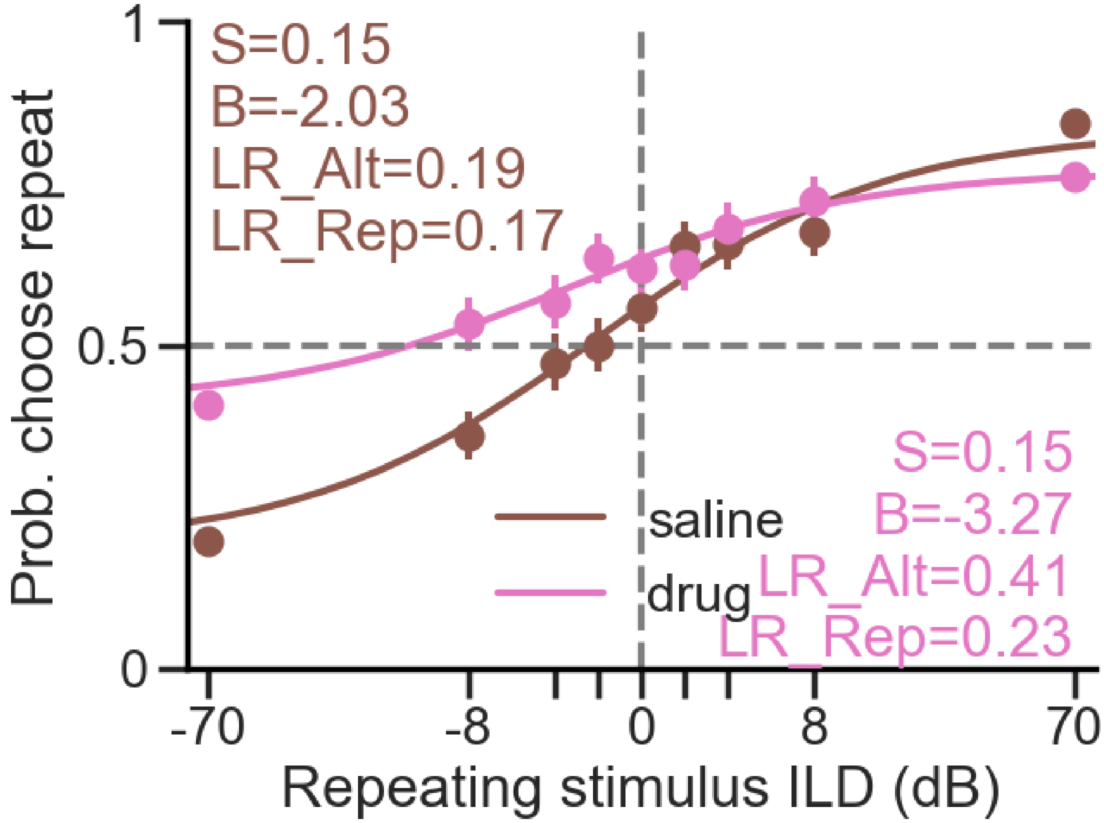


Prob. choose repeat
1
0.5
0
S=0.0
B=418.51
LR_Alt=-0.77
LR_Rep=-3.09
saline
drug
S=0.8
B=2.39
LR_Alt=0.5
LR_Rep=0.32
-70
-8
0
8
70
Repeating stimulus ILD (dB)

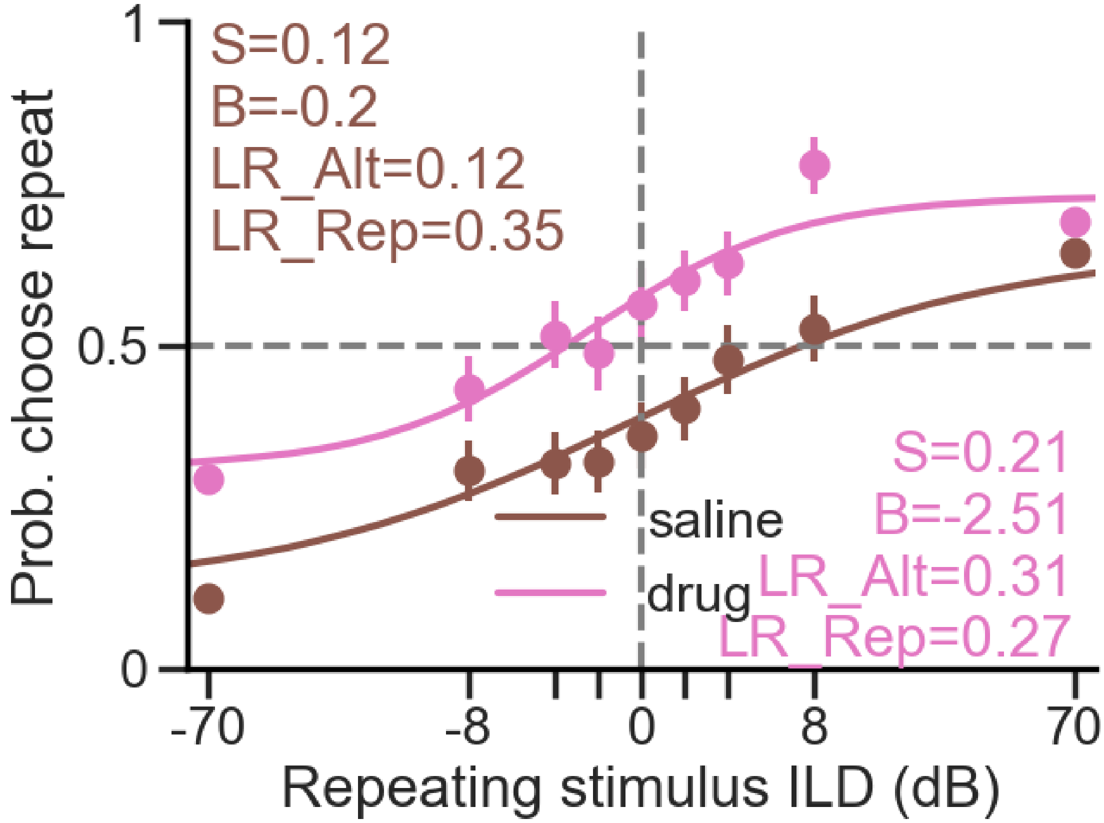


S=0.27
B=1.11
LR_Alt=0.38
LR_Rep=0.4
S=0.54
B=-4.3
LR_Alt=0.46
LR_Rep=0.39
saline
drug
Prob. choose repeat
1
0.5
0
-70
-8
0
8
70
Repeating stimulus ILD (dB)

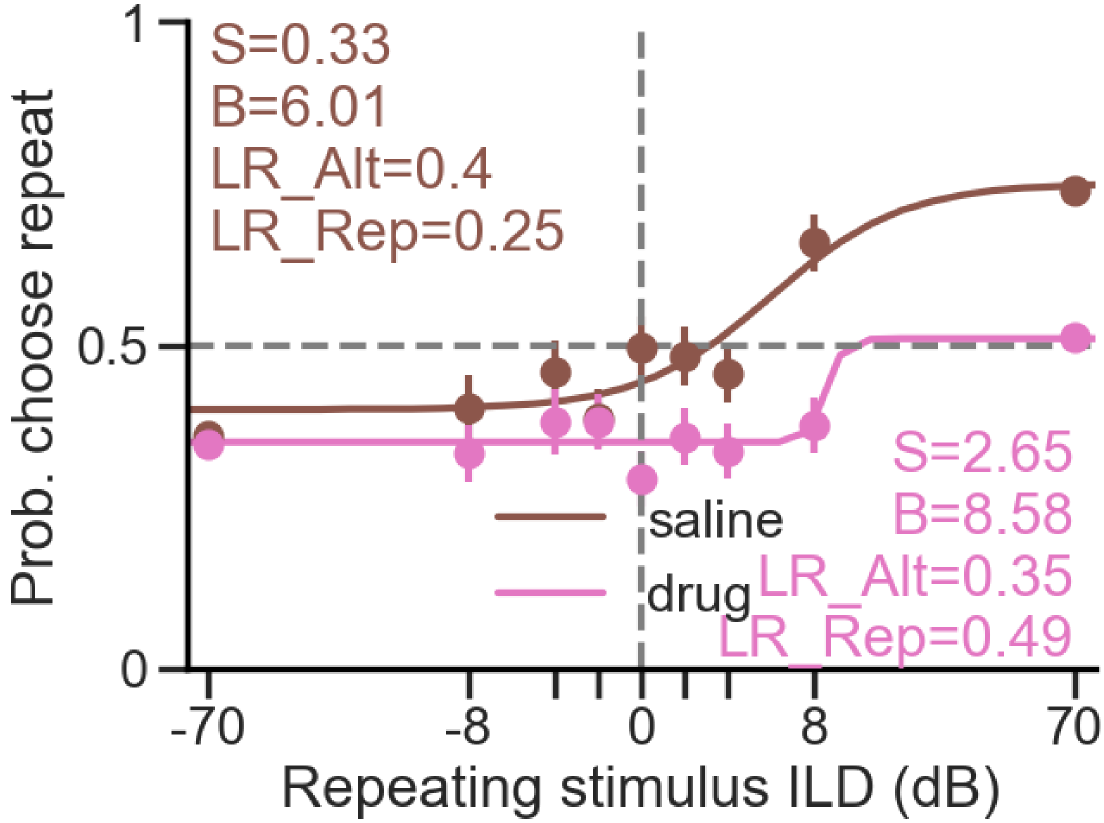

S=0.2
B=0.86
LR_Alt=0.12
LR_Rep=0.16
S=0.21
B=1.77
LR_Alt=0.37
LR_Rep=0.13
saline
drug
Prob. choose repeat
1
0.5
0
-70
-8
0
8
70
Repeating stimulus ILD (dB)

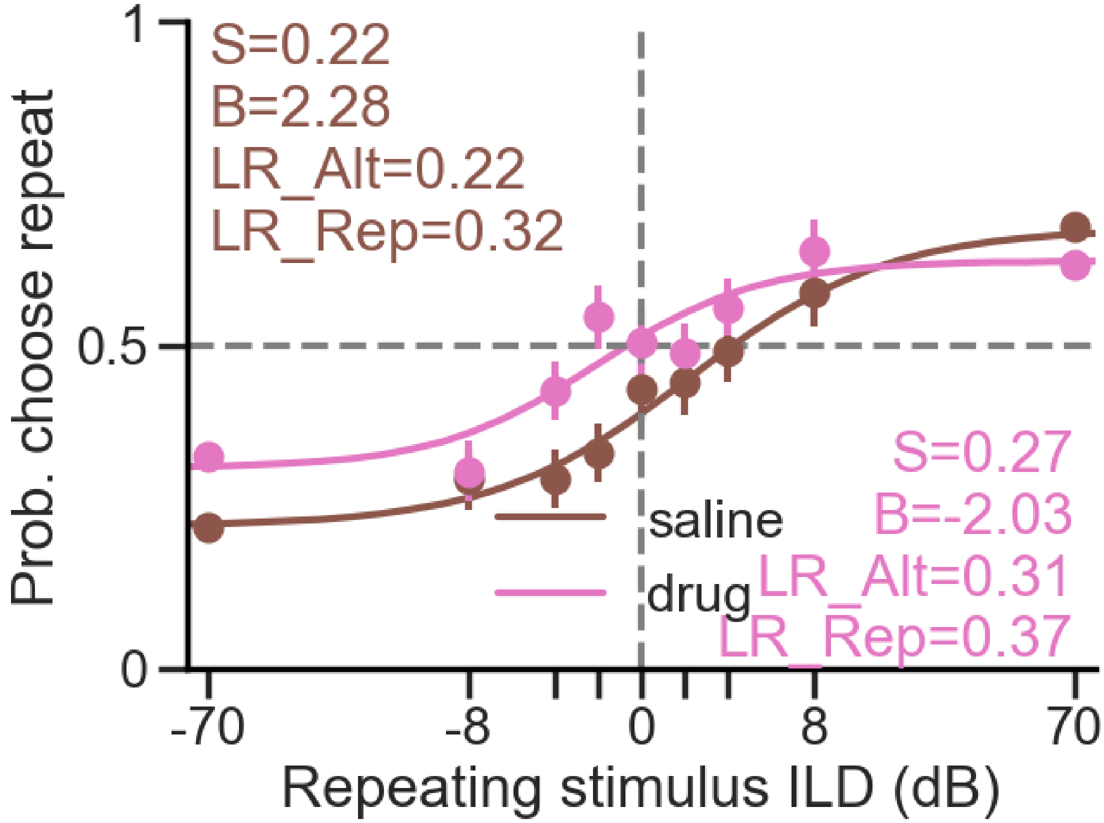


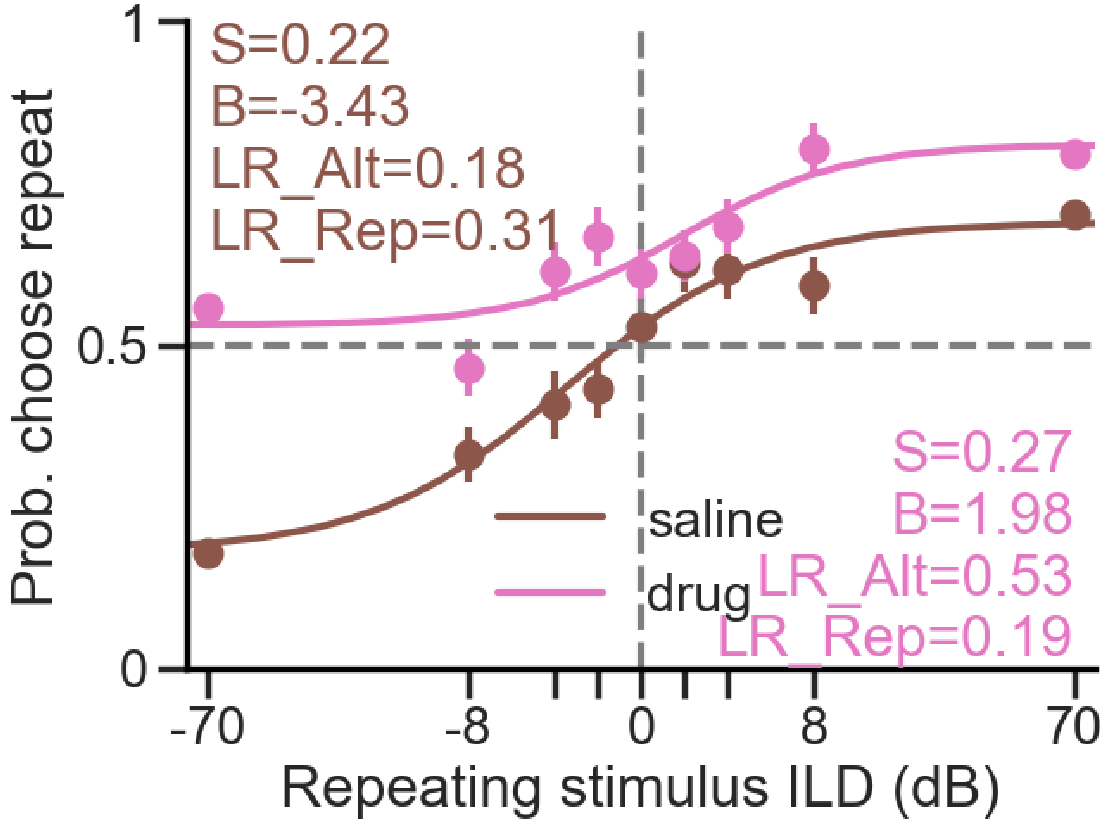
S=0.15
B=-2.03
LR_Alt=0.19
LR_Rep=0.17
S=0.15
B=-3.27
LR_Alt=0.41
LR_Rep=0.23
saline
drug
1
0.5
0
Prob. choose repeat
-70
-8
0
8
70
Repeating stimulus ILD (dB)



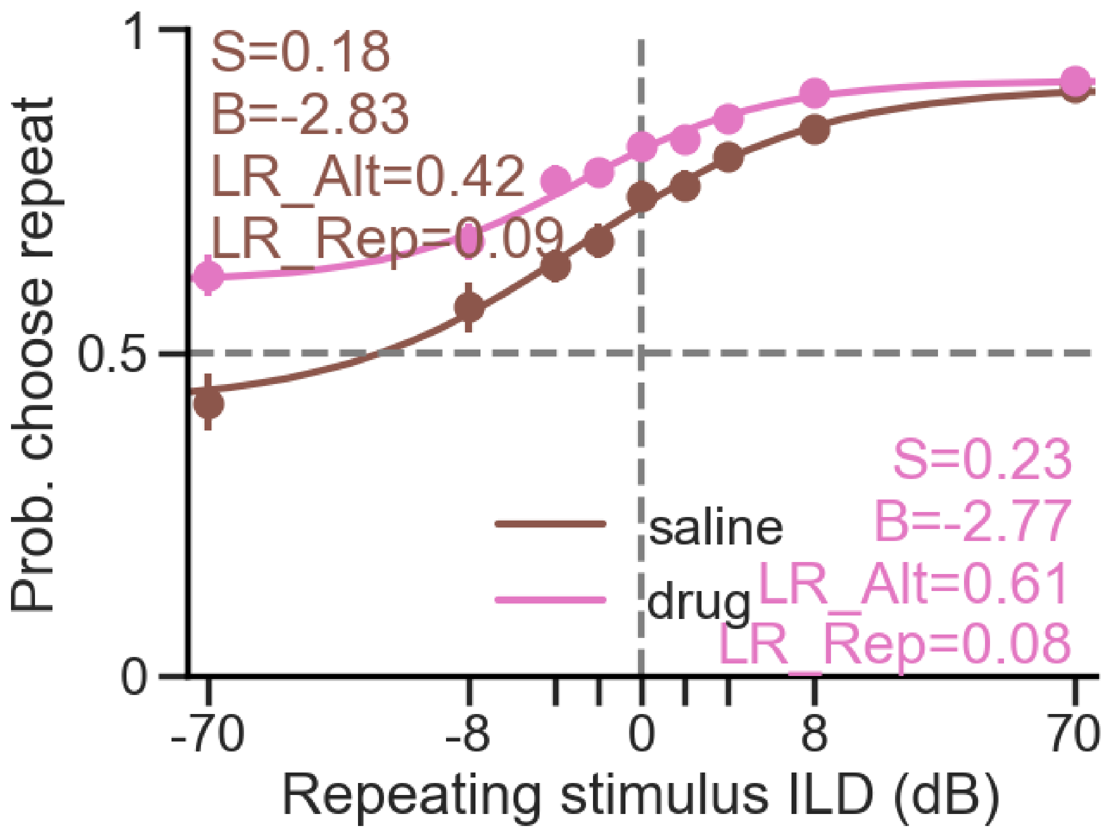
S=0.12
B=-0.2
LR_Alt=0.12
LR_Rep=0.35
S=0.21
B=-2.51
LR_Alt=0.31
LR_Rep=0.27
saline
drug
Prob. choose repeat
1
0.5
0
-70
-8
0
8
70
Repeating stimulus ILD (dB)


S=0.33
B=6.01
LR_Alt=0.4
LR_Rep=0.25
S=2.65
B=8.58
LR_Alt=0.35
LR_Rep=0.49
saline
drug
Prob. choose repeat
1
0.5
0
-70
-8
0
8
70
Repeating stimulus ILD (dB)


S=0.22
B=2.28
LR_Alt=0.22
LR_Rep=0.32
S=0.27
B=-2.03
LR_Alt=0.31
LR_Rep=0.37
saline
drug
Prob. choose repeat
1
0.5
0
-70
-8
0
8
70
Repeating stimulus ILD (dB)

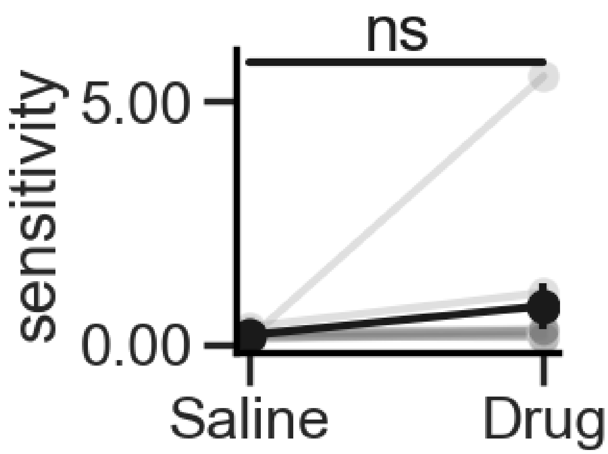

S=0.22
B=-3.43
LR_Alt=0.18
LR_Rep=0.31
S=0.27
B=1.98
LR_Alt=0.53
LR_Rep=0.19
saline
drug
Prob. choose repeat
1
0.5
0
-70
-8
0
8
70
Repeating stimulus ILD (dB)

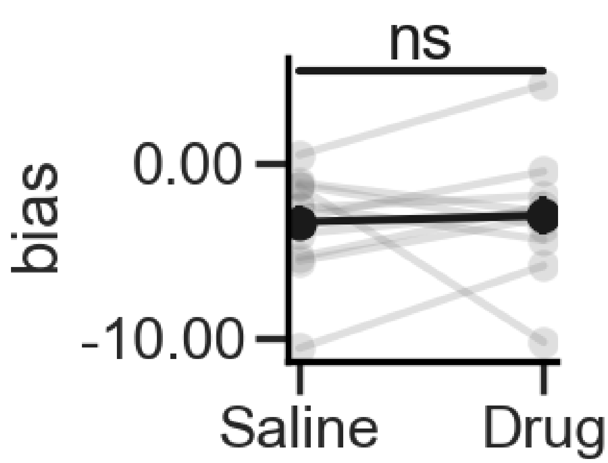
6.2.2 Mean across animals

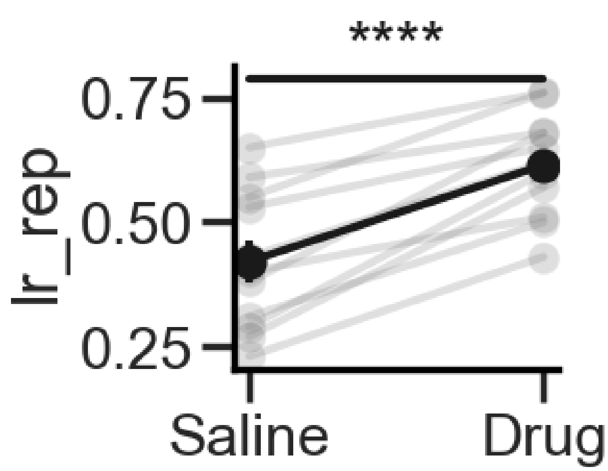


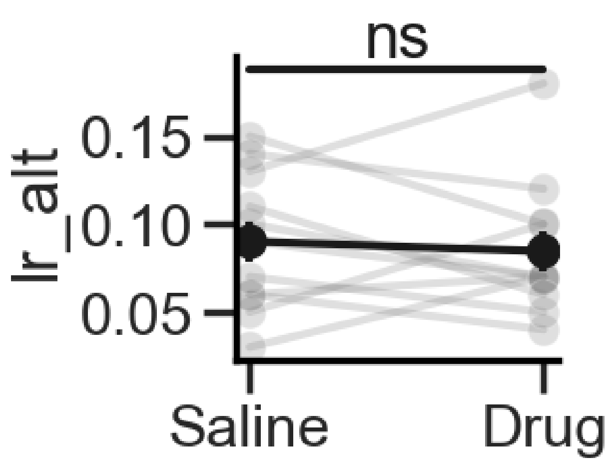
Stats

```
sensitivity: t-statistic = -1.232, p-value = 0.246
bias: t-statistic = -0.311, p-value = 0.762
lr_rep: t-statistic = -7.866, p-value = 0.000
lr_alt: t-statistic = 0.486, p-value = 0.638
```


ns
sensitivity
5.00
0.00
Saline
Drug


ns
bias
0.00
-10.00
Saline
Drug


****
lr_rep
0.75
0.50
0.25
Saline
Drug



7 Kernels

7.1 Psychophysical kernels

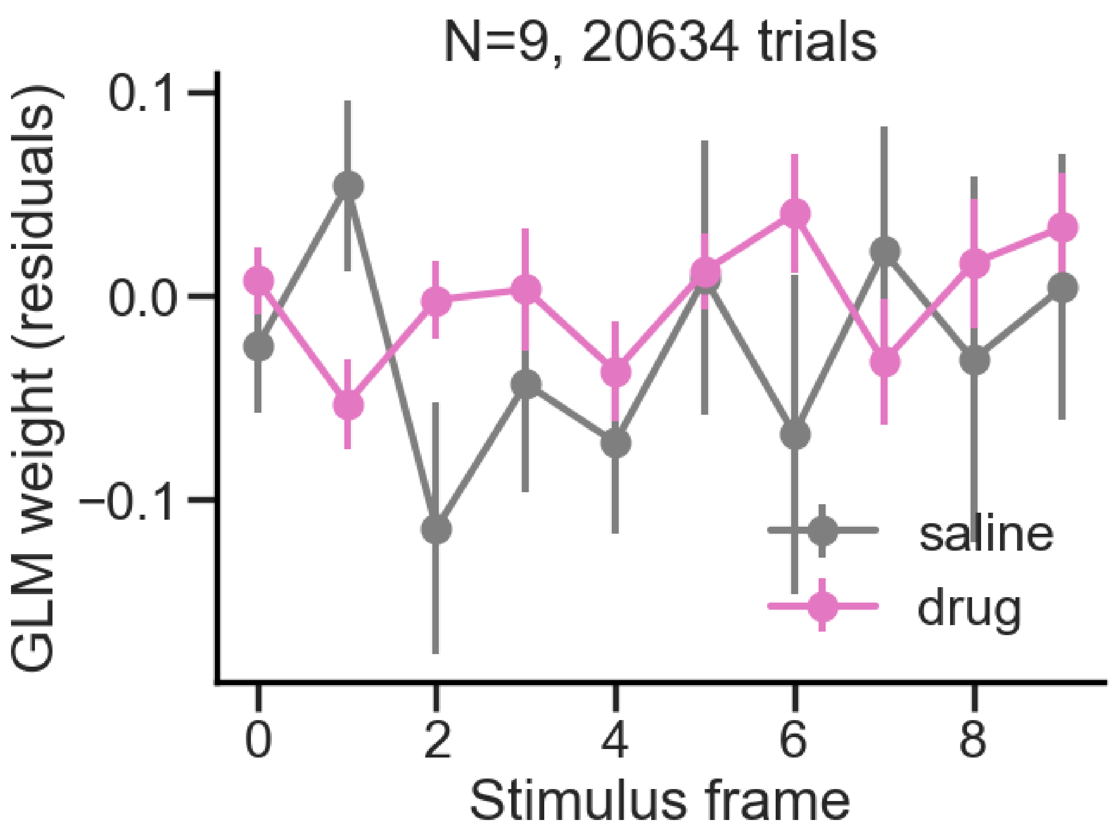
7.1.1 Mean across animals

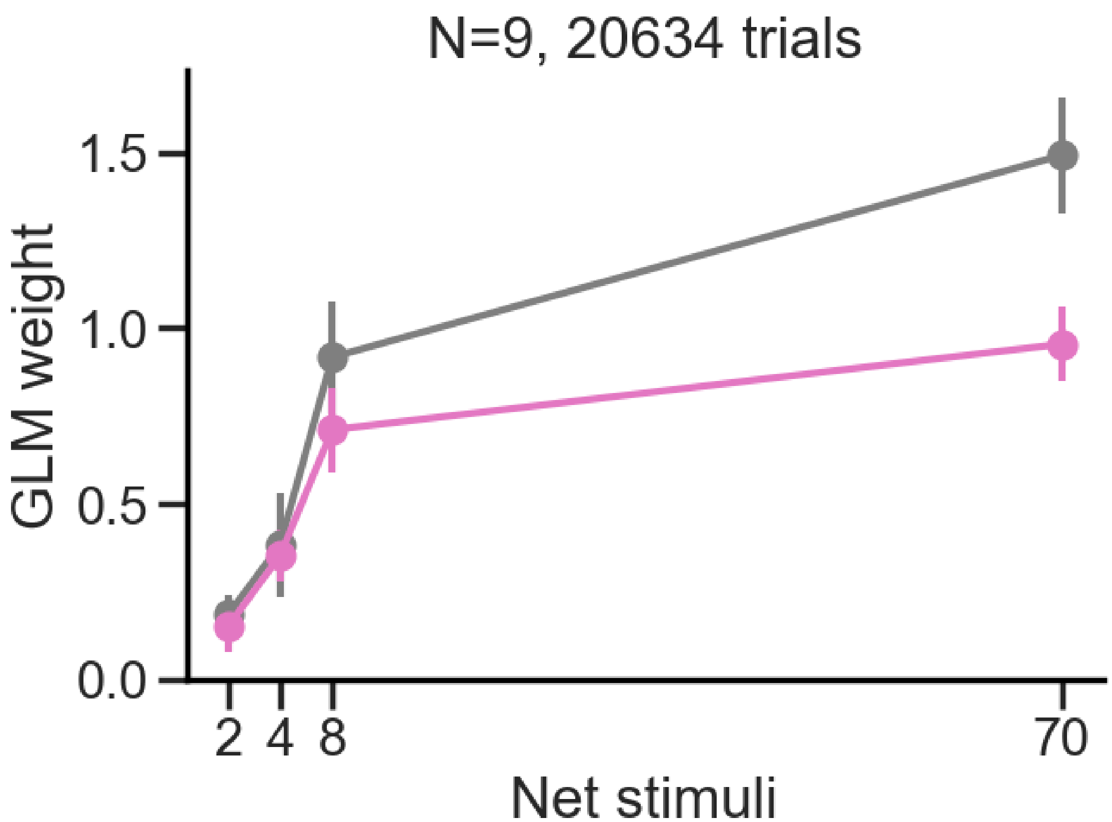
```
2AFC_6
C:\Users\Usuario\PycharmProjects\glue_sessions\2AFC_6
Getting kernel of animal 014 (1/9)
get_pk took 1.06 seconds to run
Getting kernel of animal 016 (2/9)
get_pk took 1.85 seconds to run
Getting kernel of animal 017 (3/9)
get_pk took 1.97 seconds to run
Getting kernel of animal 020 (4/9)
get_pk took 2.28 seconds to run
Getting kernel of animal 021 (5/9)
get_pk took 1.75 seconds to run
Getting kernel of animal 022 (6/9)
get_pk took 1.22 seconds to run
Getting kernel of animal 023 (7/9)
get_pk took 0.89 seconds to run
Getting kernel of animal 024 (8/9)
get_pk took 1.72 seconds to run
Getting kernel of animal 025 (9/9)
get_pk took 1.40 seconds to run
get_mean_pk took 14.15 seconds to run
2AFC_6
C:\Users\Usuario\PycharmProjects\glue_sessions\2AFC_6
Getting kernel of animal 014 (1/9)
```

```
get_pk took 1.33 seconds to run
Getting kernel of animal 016 (2/9)
get_pk took 1.81 seconds to run
Getting kernel of animal 017 (3/9)
get_pk took 1.98 seconds to run
Getting kernel of animal 020 (4/9)
get_pk took 2.32 seconds to run
Getting kernel of animal 021 (5/9)
get_pk took 1.83 seconds to run
Getting kernel of animal 022 (6/9)
get_pk took 1.46 seconds to run
Getting kernel of animal 023 (7/9)
get_pk took 1.02 seconds to run
Getting kernel of animal 024 (8/9)
get_pk took 1.73 seconds to run
Getting kernel of animal 025 (9/9)
```

```
No artists with labels found to put in legend.  Note that artists whose label
start with an underscore are ignored when legend() is called with no argument.
```

```
get_pk took 1.46 seconds to run
get_mean_pk took 14.94 seconds to run
plot_drug_pk took 29.12 seconds to run
```






```
2AFC_6
C:\Users\Usuario\PycharmProjects\glue_sessions\2AFC_6
Getting kernel of animal 014 (1/9)
get_pk took 1.05 seconds to run
Getting kernel of animal 016 (2/9)
get_pk took 1.73 seconds to run
Getting kernel of animal 017 (3/9)
get_pk took 1.98 seconds to run
Getting kernel of animal 020 (4/9)
get_pk took 2.24 seconds to run
Getting kernel of animal 021 (5/9)
get_pk took 1.72 seconds to run
Getting kernel of animal 022 (6/9)
get_pk took 1.23 seconds to run
Getting kernel of animal 023 (7/9)
get_pk took 0.89 seconds to run
Getting kernel of animal 024 (8/9)
get_pk took 1.65 seconds to run
Getting kernel of animal 025 (9/9)
```

```
get_pk took 1.38 seconds to run
get_mean_pk took 13.87 seconds to run
2AFC_6
C:\Users\Usuario\PycharmProjects\glue_sessions\2AFC_6
Getting kernel of animal 014 (1/9)
get_pk took 1.31 seconds to run
Getting kernel of animal 016 (2/9)
get_pk took 1.84 seconds to run
Getting kernel of animal 017 (3/9)
```

7.2 History kernels

8 Serial dependence

Serial dependence requires a continuous readout of behavioral responses. As choices are binary (left/right), other readouts are needed. Let's try with reaction times (RT) and lick rate (nLicks)

8.0.1 Measured as Reaction Times (RTs)

ΔILD

Absolute ΔILD

8.0.2 Measured as N licks

ΔILD

Absolute ΔILD

9 Save all notebook's figures